SECOND LAW OF THERMODYNAMICS

OBJECTIVES

Introduce the ***second law of thermodynamics***.

Identify valid processes as those that satisfy both the first and second laws of thermodynamics.

Discuss thermal energy reservoirs, ***reversible*** and ***irreversible processes, heat engines, refrigerators, and heat pumps***.

Describe ***the Kelvin–Planck and Clausius statements of the second law of thermodynamics.***

Discuss the ***concepts of perpetual-motion*** machines.

Apply the second law of thermodynamics to ***cycles and cyclic devices***.

Apply the second law to ***develop the absolute thermodynamic temperature scale.***

Describe the ***Carnot cycle and carnot theorems***

Examine the Carnot principles, idealized Carnot heat engines, refrigerators, and heat pumps.

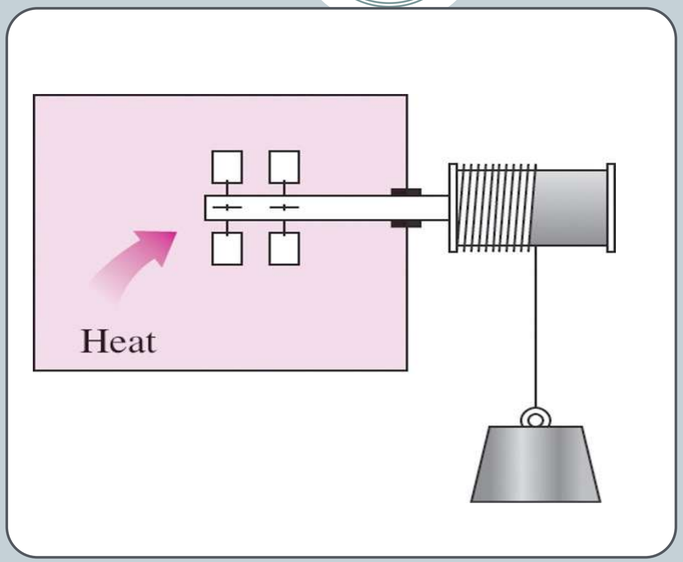
Determine the expressions for the thermal efficiencies and coefficients of performance for reversible heat engines, heat pumps, and refrigerators.

INTRODUCTION TO THE SECOND LAW

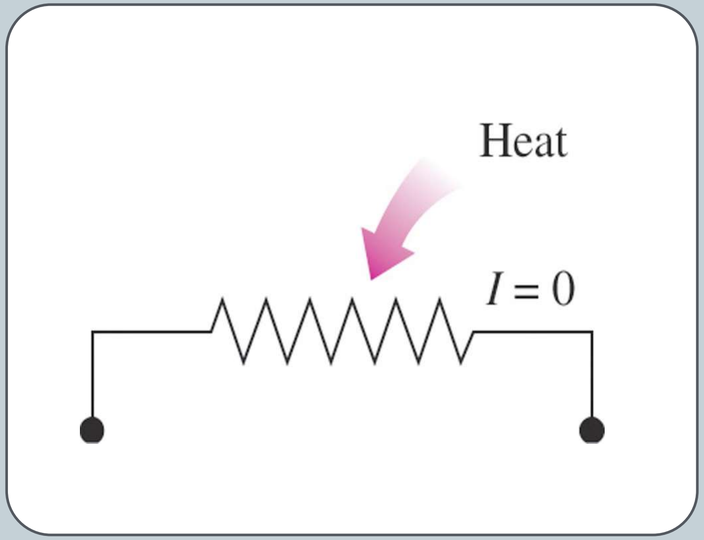
These processes cannot occur even though they are not in violation of the first law.



A cup of hot coffee does not get hotter in a cooler room.



Transferring heat to a paddle wheel will not cause it to rotate.



Transferring heat to a wire will not generate electricity.

PROCESSES

occur in a certain direction & not in the reverse direction

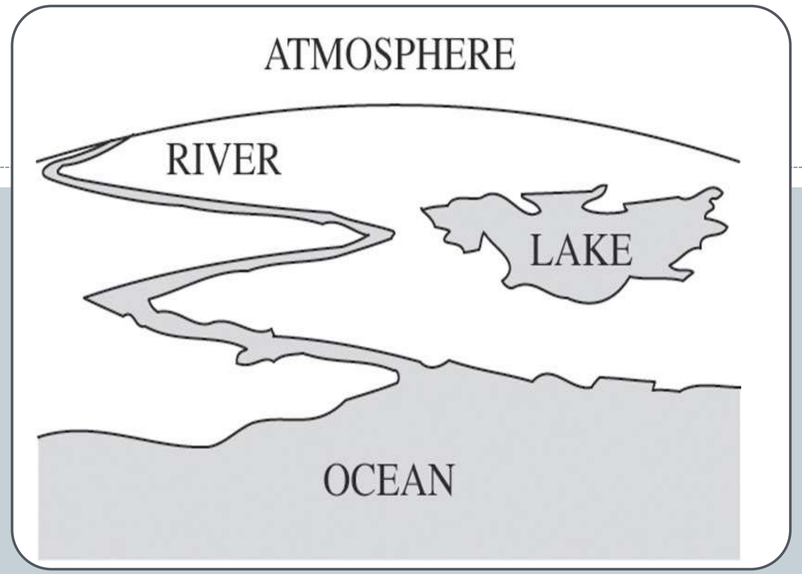
USES OF SECOND LAW OF THERMODYNAMICS

The second law may be used to ***identify the direction of processes***.

The second law also asserts that energy has *quality* as well as quantity. The first law is concerned with the quantity of energy and the transformations of energy from one form to another with no regard to its quality. The second law provides the necessary means to ***determine the quality as well as the degree of degradation of energy during a process***.

The second law of thermodynamics is also used ***in determining the theoretical limits for the performance of commonly used engineering systems***, such as heat engines and refrigerators, as well as predicting the *degree of completion* of chemical reactions.

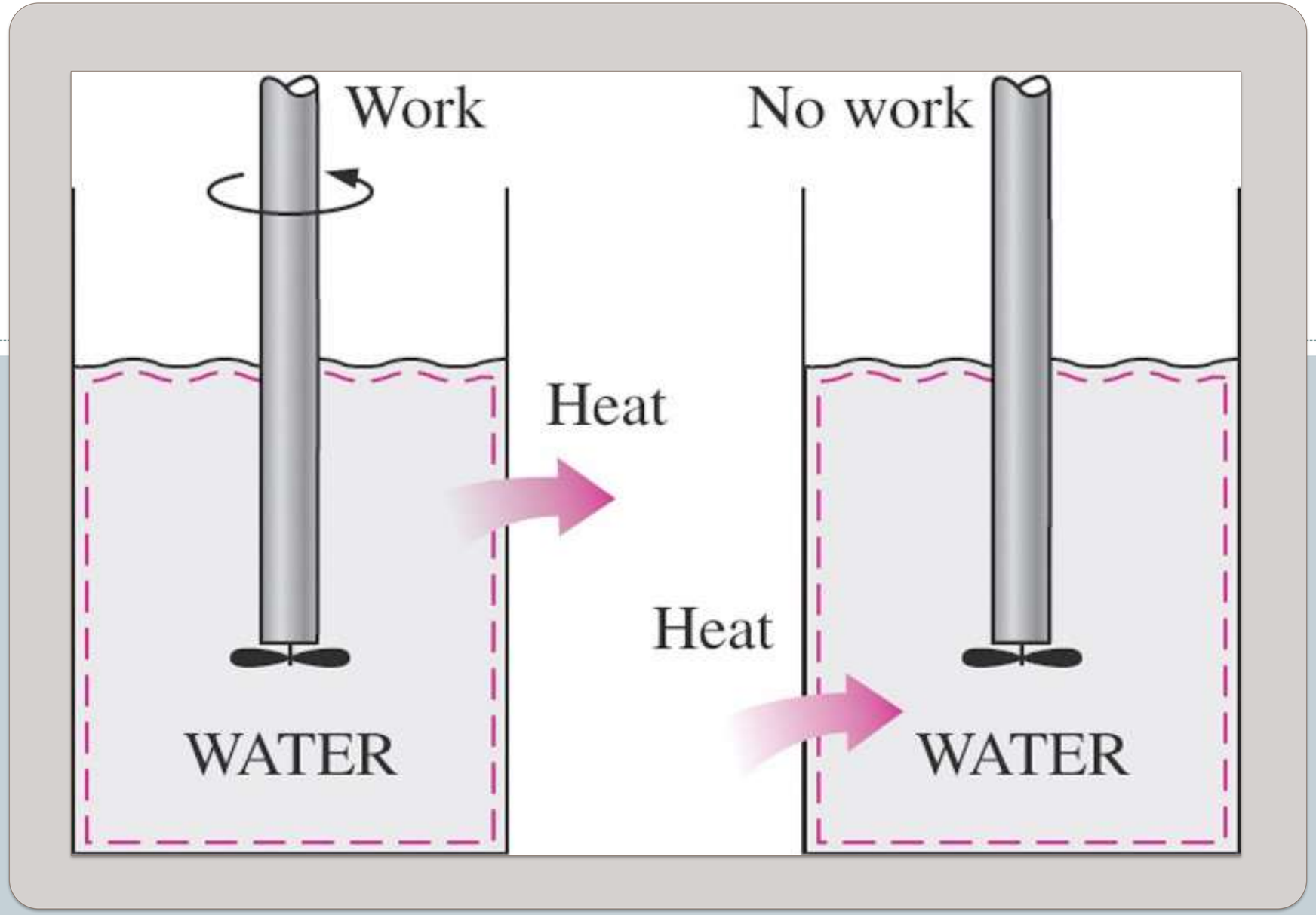
THERMAL ENERGY RESERVOIRS



Bodies with relatively large thermal masses can be modelled as thermal energy reservoirs.

A ***thermal energy reservoir (TER)*** is defined as a large body of infinite heat capacity, which is capable of absorbing or rejecting an unlimited quantity of heat without suffering appreciable changes in its thermodynamic coordinates.

In practice, large bodies of water such as oceans, lakes, and rivers as well as the atmospheric air can be modelled accurately as thermal energy reservoirs because of their large thermal energy storage capabilities or thermal masses.



Work can always be converted to heat directly and completely, but the reverse is not true

Differences Between Work and Heat as High-Grade and Low-Grade Energy

Work(high grade energy)

- Work is a high-grade form of energy.
- Work is an organized mode of energy transfer.
- Work can be ***completely converted into heat.***
- Work can be ***converted completely into useful work*** in an ideal process.
- Work has a high capacity to produce useful work.

Heat(low grade energy)

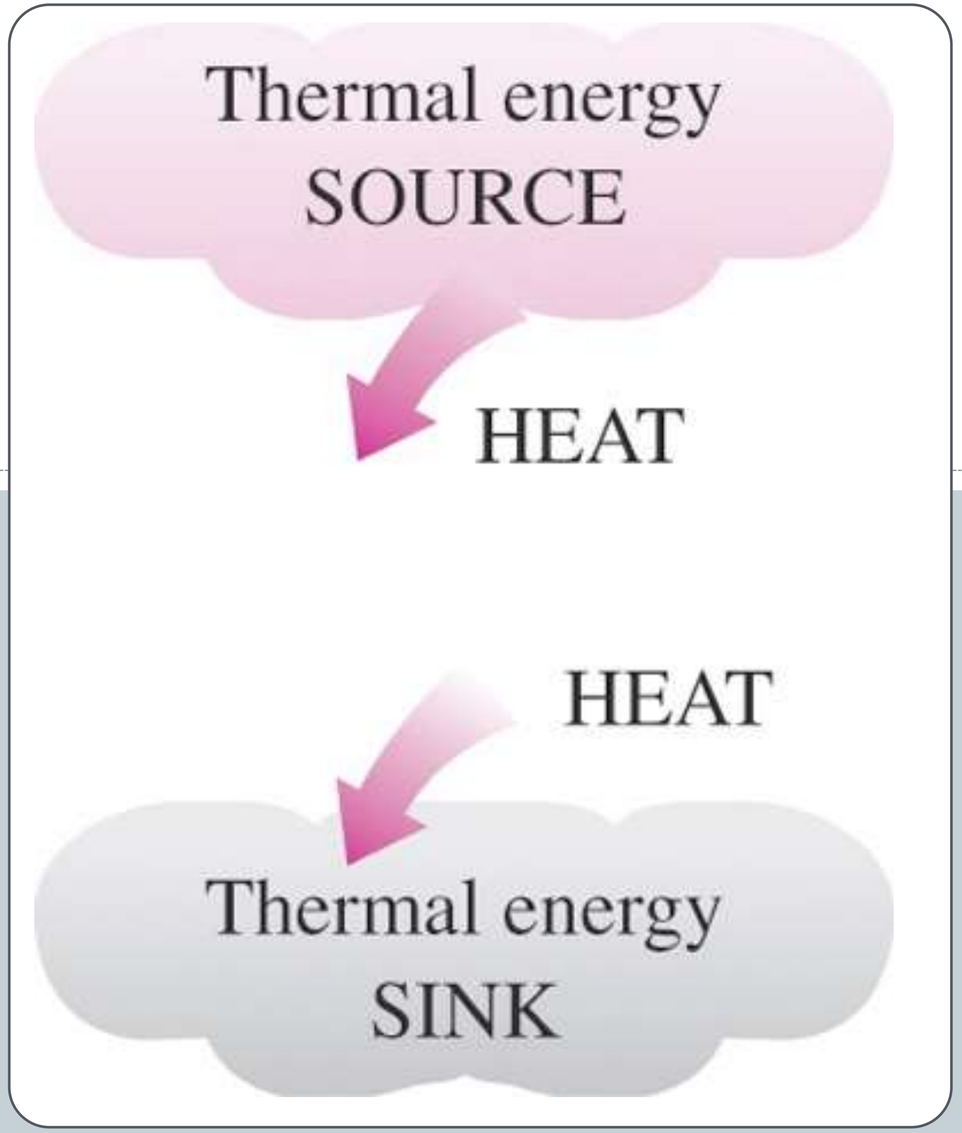
- Heat is generally considered a low-grade form of energy.
- Heat transfer occurs due to a temperature difference.
- ***Heat cannot be completely converted into work in a cyclic process***.
- ***Only a part of heat can be converted into work in a cyclic heat engine***.
- The capacity of heat to produce useful work depends on its temperature and the surroundings.

HEAT ENGINES

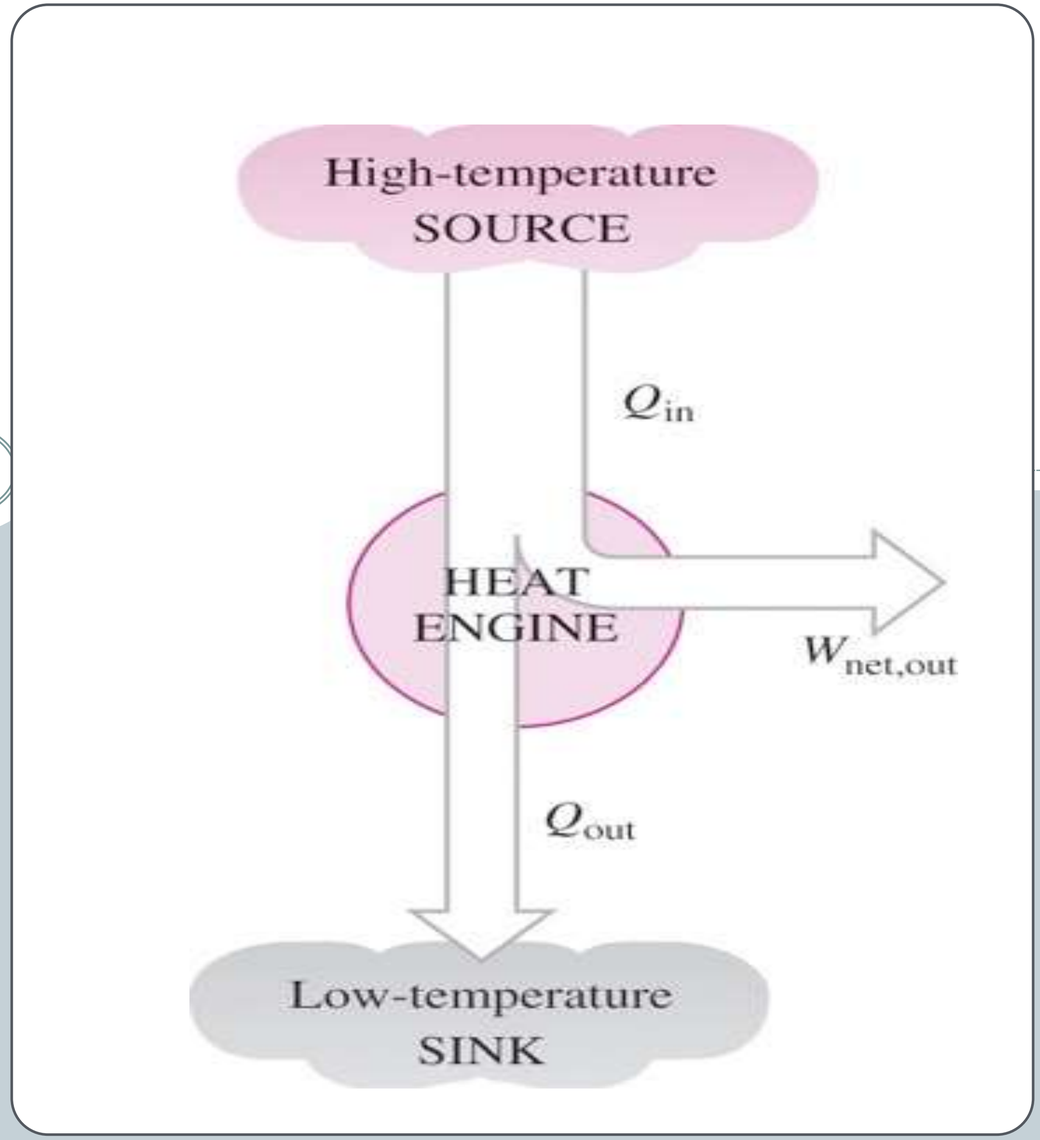
A ***heat engine*** may be defined as a device which operates in a thermodynamic cycle, converts part of the heat received from a high-temperature reservoir to useful work and rejects the remaining heat to a low-temperature reservoir

- They receive heat from a high-temperature source (solar energy, oil furnace, nuclear reactor, etc.)
- They convert part of this heat to work (usually in the form of a rotating shaft.)
- They reject the remaining waste heat to a low-temperature sink (the atmosphere, rivers, etc.)
- They operate on a cycle

Heat engines and other cyclic devices usually involve a fluid to and from which heat is transferred while undergoing a cycle. This fluid is called the ***working fluid***

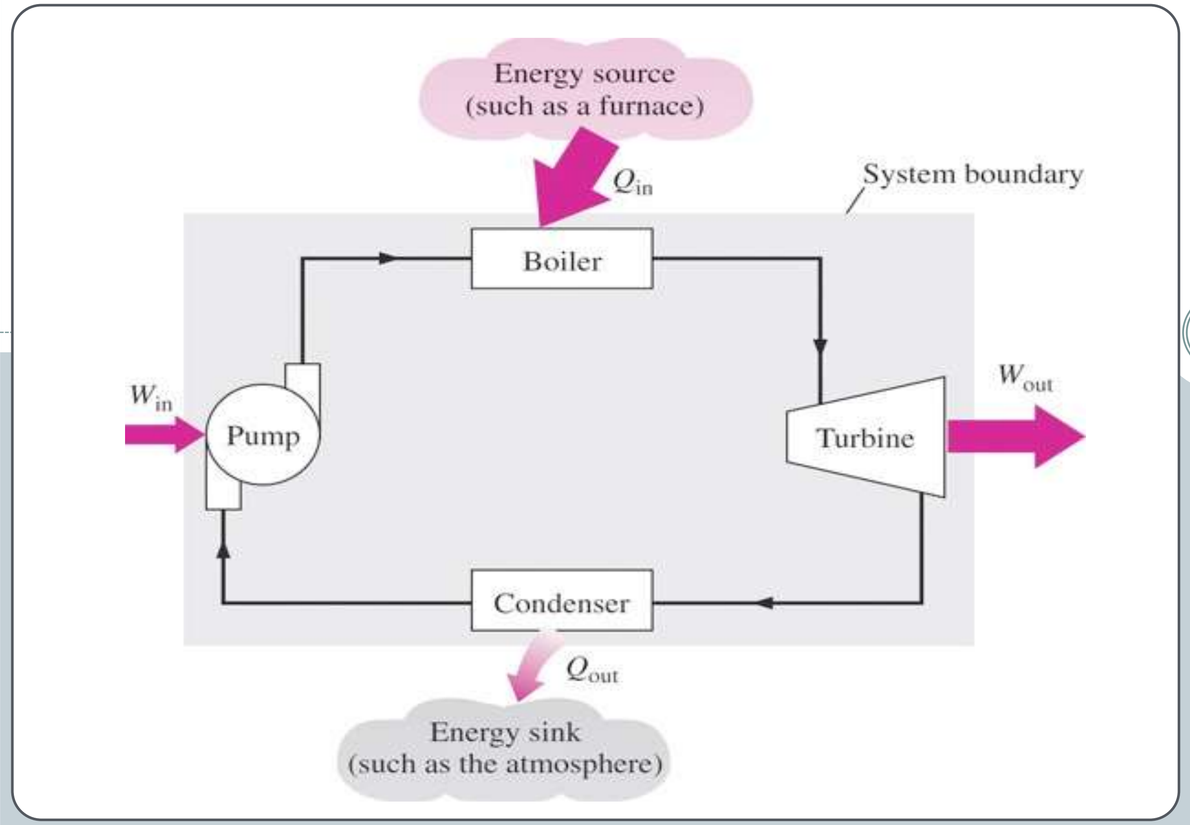


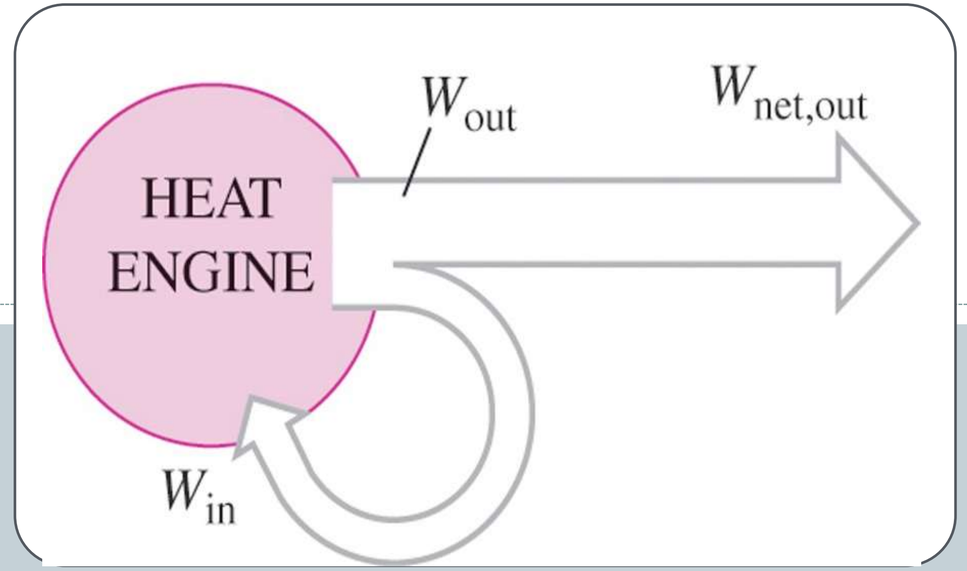
A source supplies energy in the form of heat and a sink absorbs it



Part of the heat received by a heat engine is converted to work, while the rest is rejected to a sink

Steam power plant





A portion of the work output of a heat engine is consumed internally to maintain continuous operation.

$$W_{net,\,out} = W_{out} - W_{in} \quad (kJ)$$

$$W_{net,\,out} = Q_{in} - Q_{out} \quad (kJ)$$

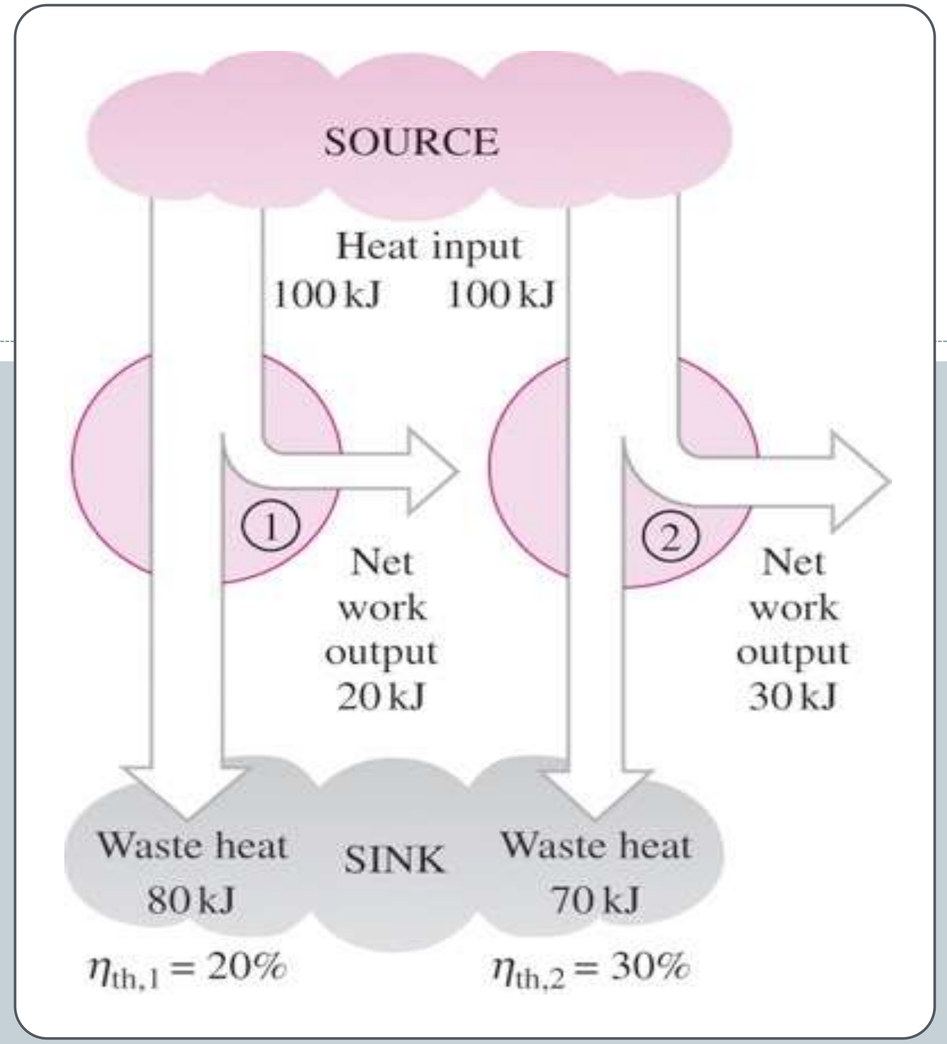
Q_{in} = amount of heat supplied to steam in boiler from a high temperature source (furnace)

Q_{out} = amount of heat rejected from steam in condenser to a low-temperature sink (the atmosphere, a river, etc...)

W_{in} = amount of work required to compress water to boiler pressure

W_{out} = amount of work delivered by steam as it expands in turbine

Thermal efficiency



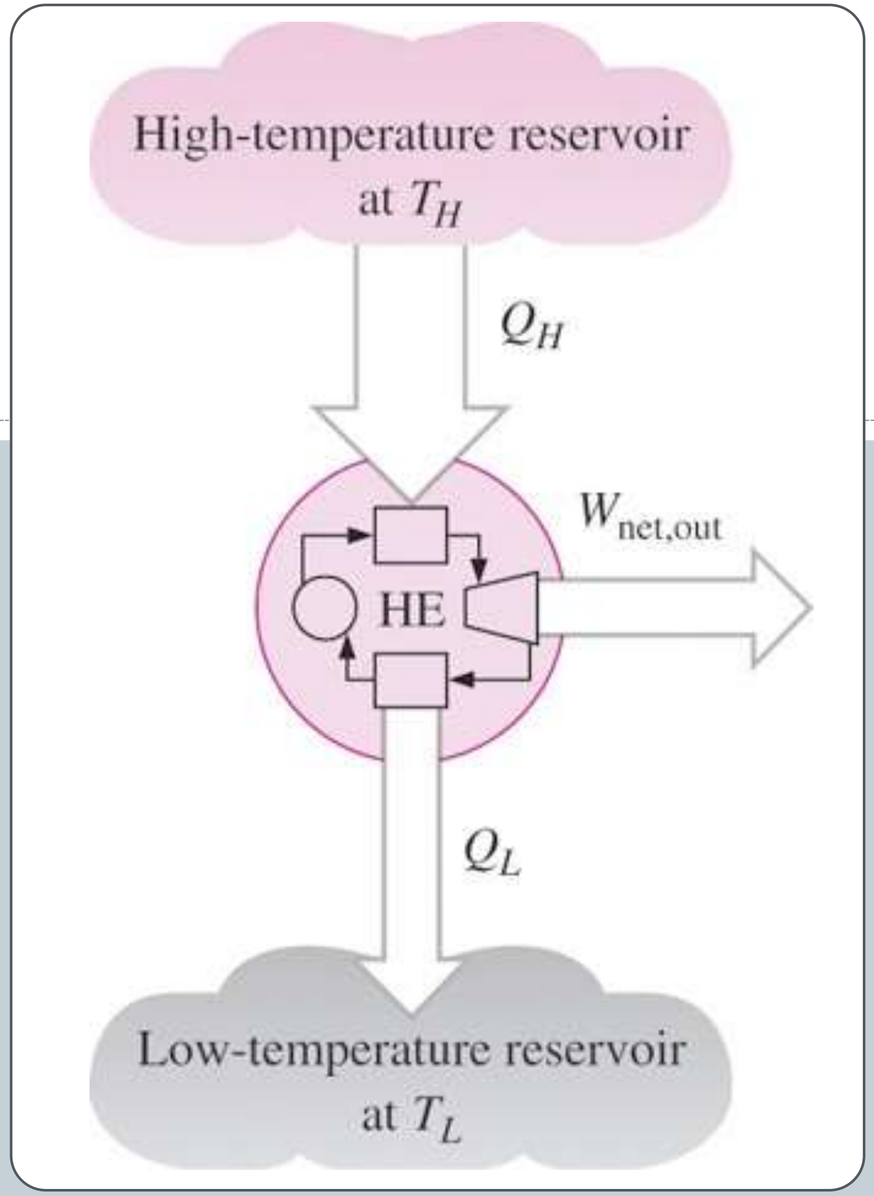
Some heat engines perform better than others (convert more of the heat they receive to work).

$$\text{Thermal efficiency} = \frac{\text{Net work output}}{\text{Total heat input}}$$

$$\eta_{th} = \frac{W_{net,\,out}}{Q_{in}}$$

$$\eta_{th} = 1 - \frac{Q_{out}}{Q_{in}}$$

$$W_{net,\,out} = Q_{in} - Q_{out}$$

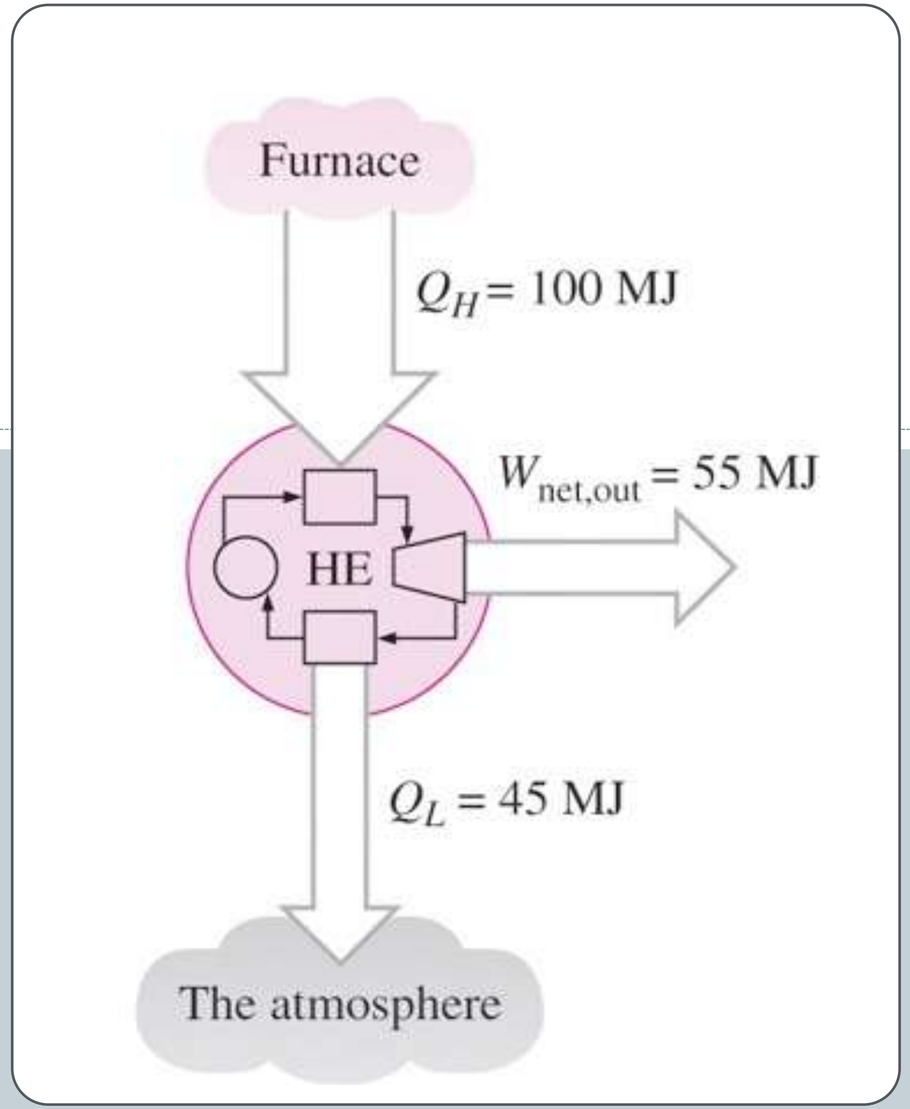


Schematic of a heat engine

$$W_{net,\,out} = Q_H - Q_L$$

$$\eta_{th} = \frac{W_{net,\,out}}{Q_H}$$

$$\eta_{th} = 1 - \frac{Q_L}{Q_H}$$



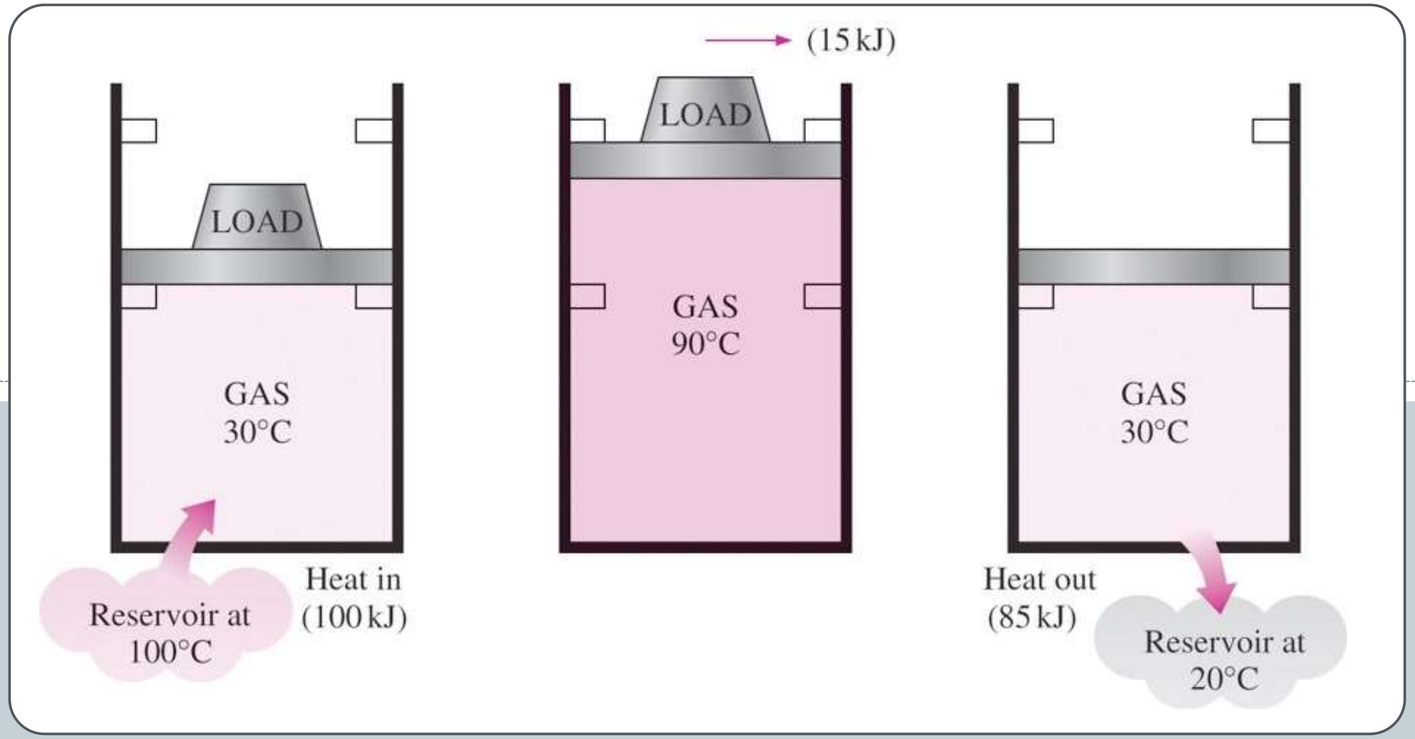
Even the most efficient heat engines reject almost one-half of the energy they receive as waste heat

Can we save Q_{out}?

In a steam power plant the condenser is the device where large quantities of waste heat is rejected to rivers, lakes, or the atmosphere.

Can we not just take the condenser out of the plant and save all that waste energy?

No - without a heat rejection process in a condenser the cycle cannot be completed.



A heat-engine cycle cannot be completed without rejecting some heat to a low-temperature sink.

Every heat engine must ***waste/release*** some energy by transferring it to a low-temperature reservoir in order to complete the cycle, even under idealised conditions.

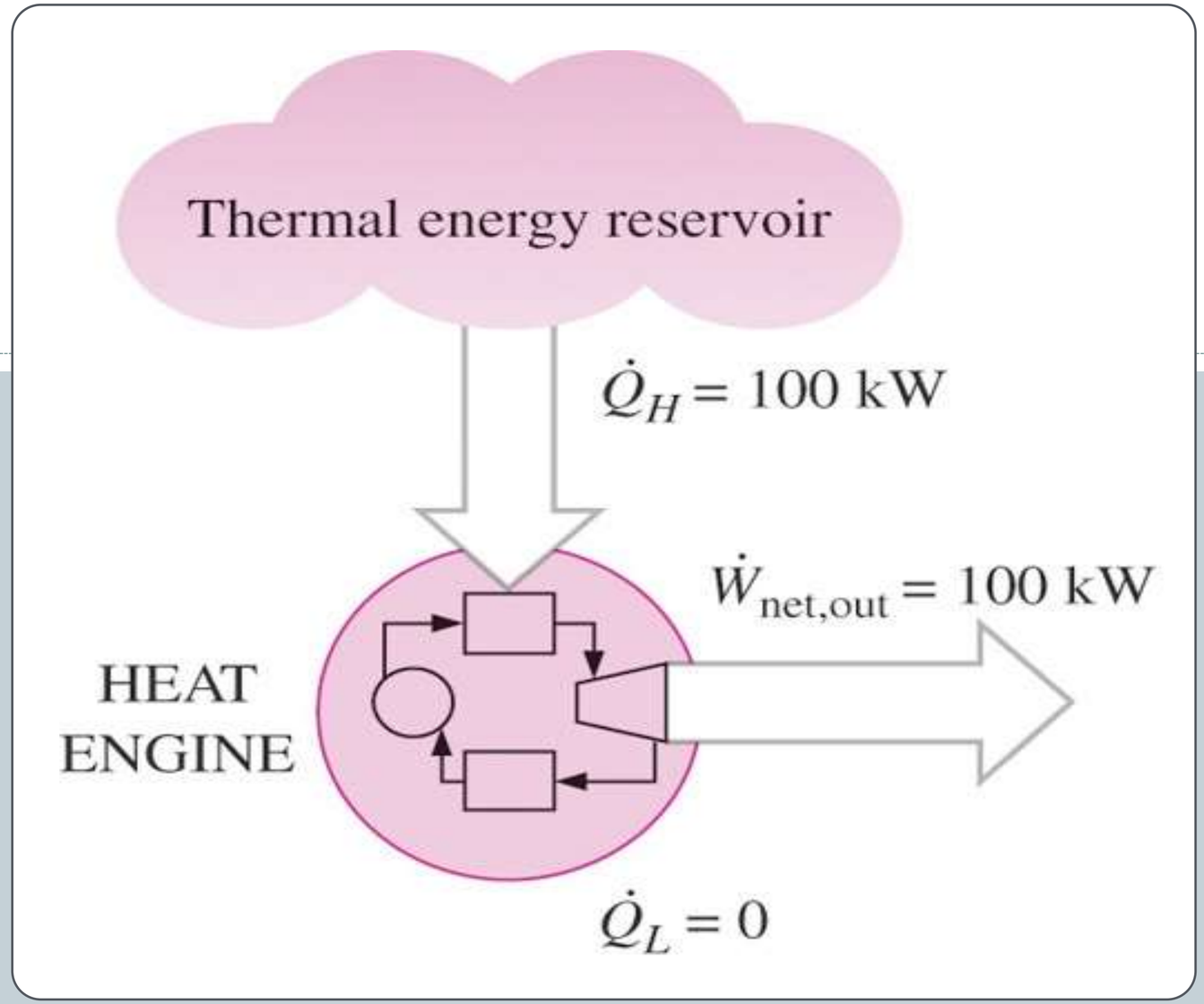
The second law of thermodynamics: KELVIN PLANCK STATEMENT

:

It is impossible for a heat engine to produce net work in a complete cycle if it exchanges heat only with bodies at a single fixed temperature.

No heat engine can have a thermal efficiency of 100 percent, or as for a power plant to operate, the working fluid must exchange heat with the environment as well as the furnace.

The impossibility of having a 100% efficient heat engine is not due to friction or other dissipative effects. It is a limitation that applies to both the idealised and the actual heat engines.



A heat engine that violates the Kelvin–Planck statement of the second law.

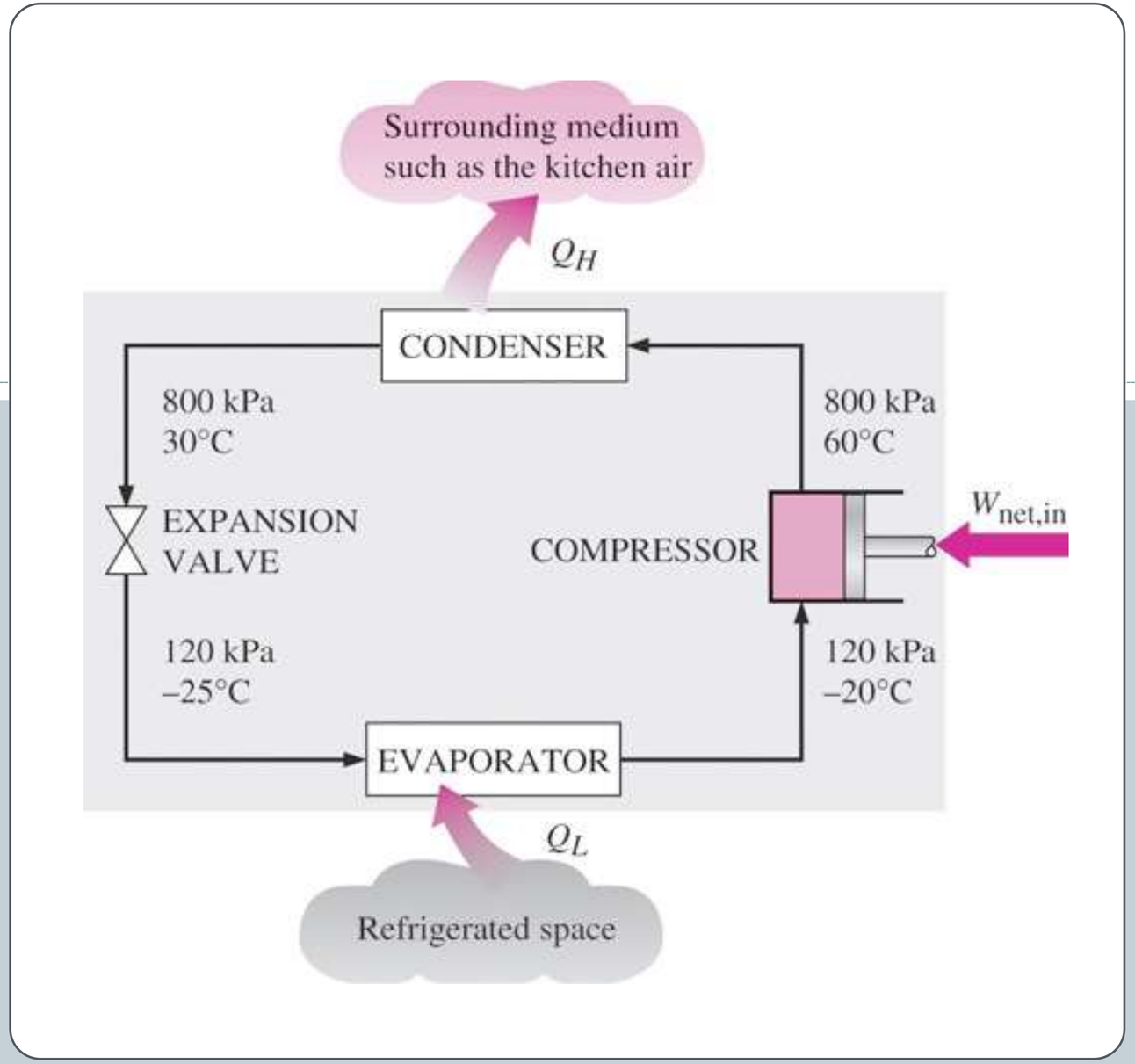
REFRIGERATORS & HEAT PUMPS

The transfer of heat from a low-temperature medium to a high-temperature one requires special devices called ***refrigerators***.

Refrigerators, like heat engines, are cyclic devices.

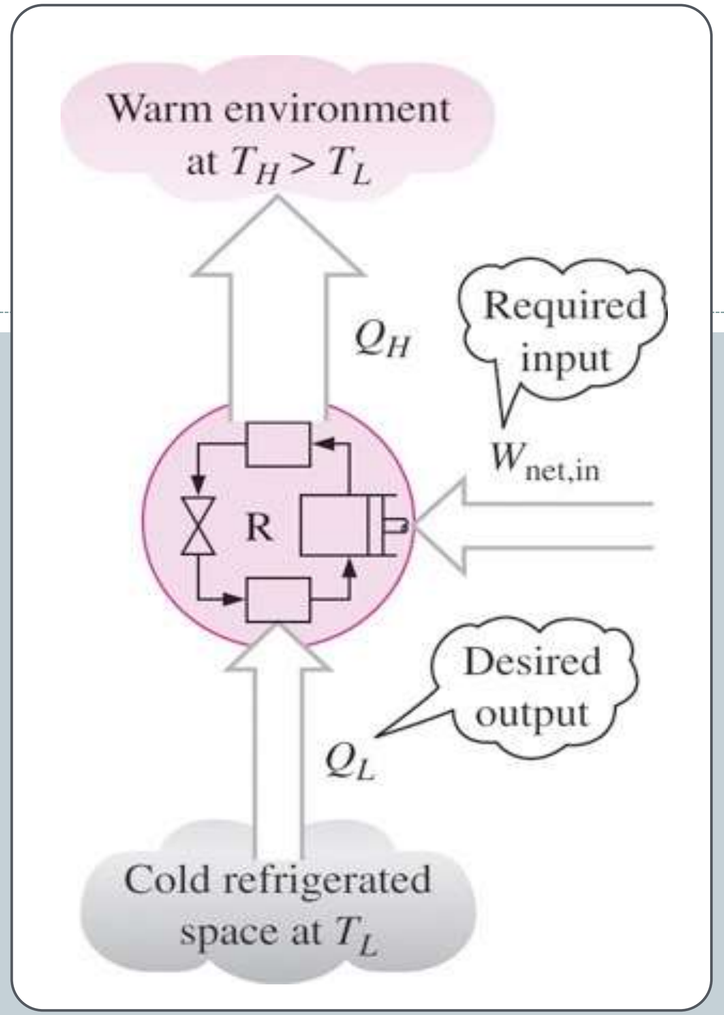
The working fluid used in the refrigeration cycle is called a ***refrigerant***.

The most frequently used refrigeration cycle is the ***vapor-compression refrigeration cycle***.



Basic components of a refrigeration system and typical operating conditions.

Coefficient of Performance



The objective of a refrigerator is to remove Q_L from the cooled space.

The ***efficiency*** of a refrigerator is expressed in terms of the ***coefficient of performance*** (COP).

The objective of a refrigerator is to ***remove heat*** (Q_L) from the refrigerated space.

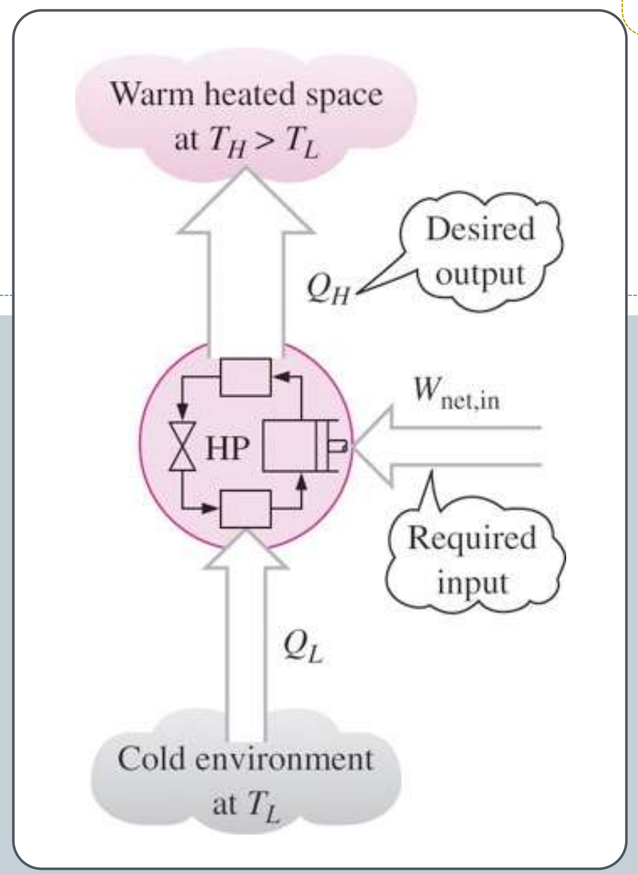
$$\text{COP}_R = \frac{\text{Desired output}}{\text{Required input}} = \frac{Q_L}{W_{net,\,in}}$$

$$W_{net,\,out} = Q_{in} - Q_{out}$$

$$\text{COP}_R = \frac{Q_L}{Q_H - Q_L} = \frac{1}{\dfrac{Q_H}{Q_L} - 1}$$

Heat Pumps

The objective of a heat pump is ***to supply heat Q_H into the warmer space.***



The objective of a heat pump is to supply heat Q_H into the warmer space.

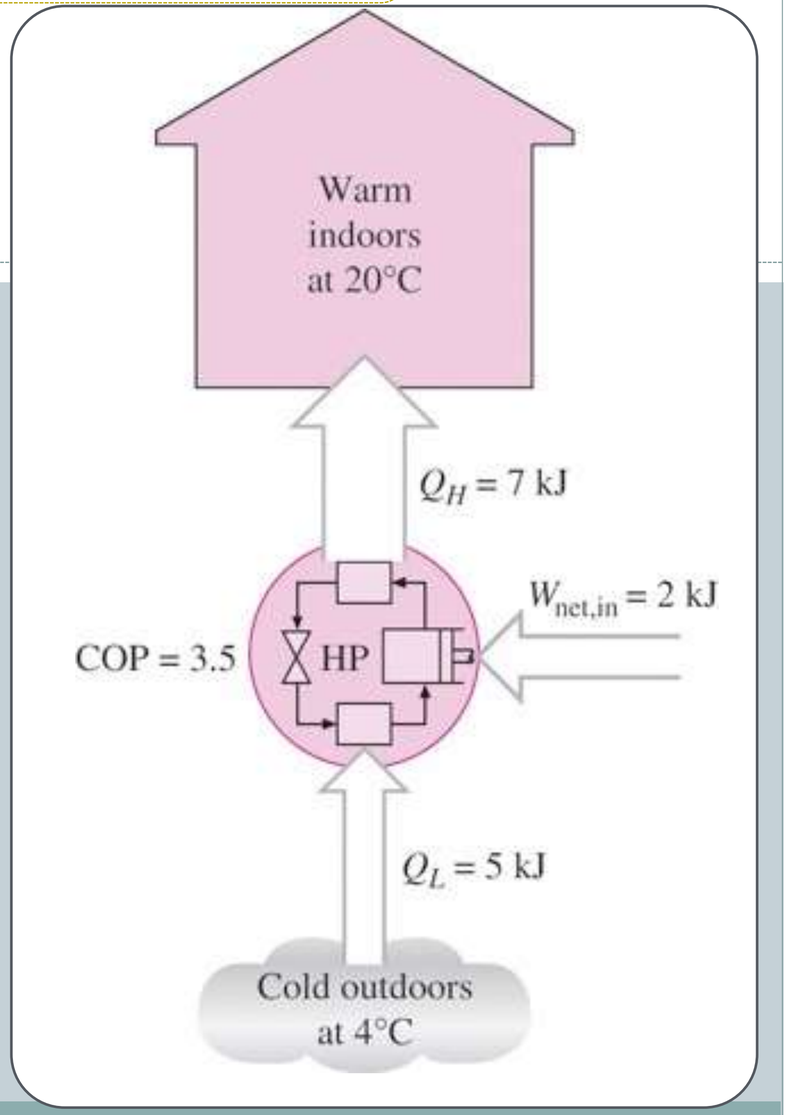
$$\text{COP}_{HP} = \frac{\text{Desired output}}{\text{Required input}} = \frac{Q_H}{W_{net,\,in}}$$

$$\text{COP}_{HP} = \frac{Q_H}{Q_H - Q_L} = \frac{1}{1 - Q_L/Q_H}$$

$$\text{COP}_{HP} = \text{COP}_R + 1$$

for fixed values of Q_L and Q_H

The work supplied to a heat pump is used to extract energy from the cold outdoors and carry it into the warm indoors.



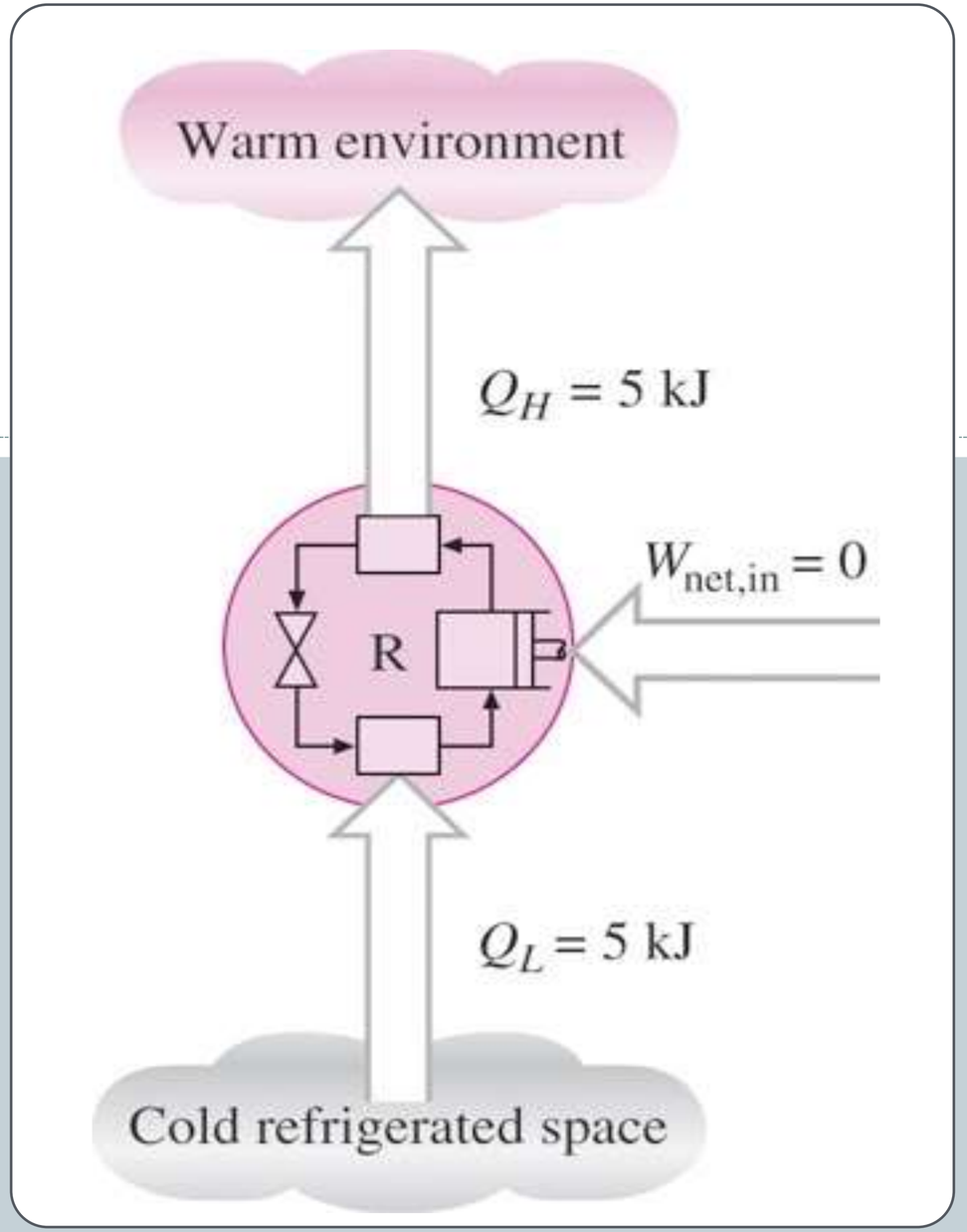
The second law of thermodynamics: Clausius statement

It is impossible to construct a device that operates in a cycle and produces no effect other than the transfer of heat from a lower-temperature body to a higher-temperature body.

It states that a refrigerator cannot operate unless its compressor is driven by an external power source, such as an electric motor.

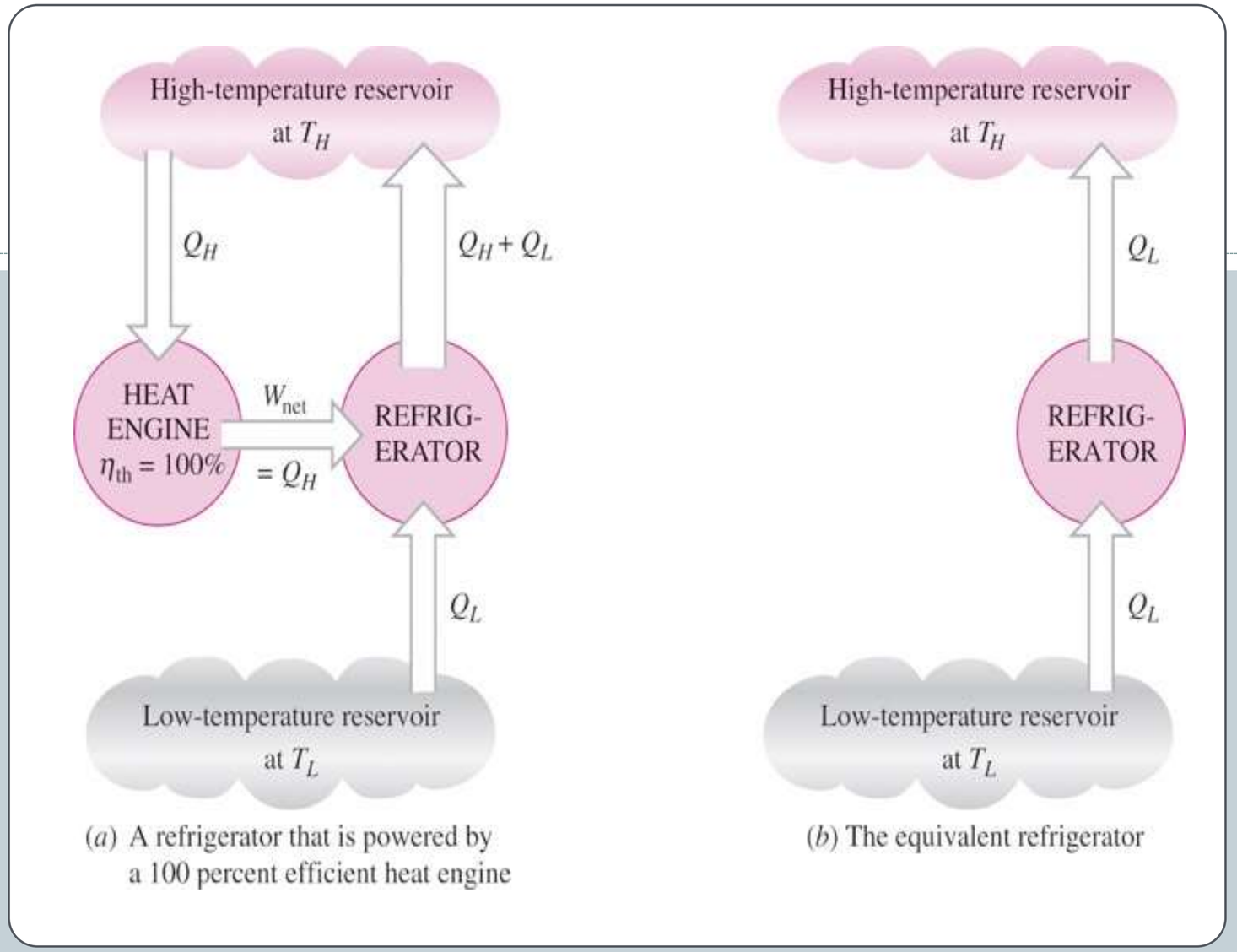
This way, the net effect on the surroundings involves the consumption of some energy in the form of work, in addition to the transfer of heat from a colder body to a warmer one.

To date, no experiment has been conducted that contradicts the second law, and this should be taken as sufficient proof of its validity.



A refrigerator that violates the Clausius statement of the second law.

Equivalence of the two statements



(*a*) A refrigerator that is powered by a 100 percent efficient heat engine

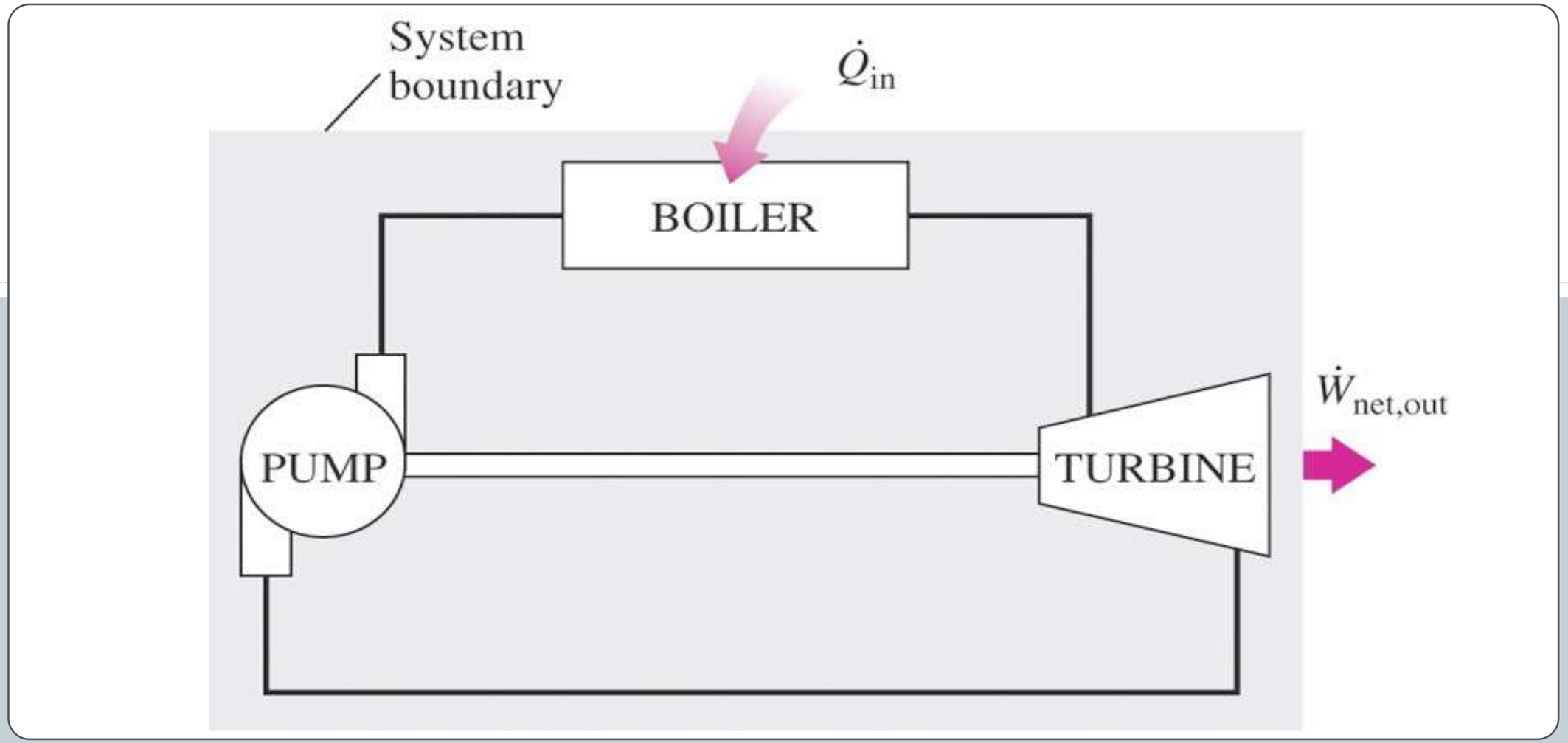
(*b*) The equivalent refrigerator

Proof that the violation of the Kelvin–Planck statement leads to the violation of the Clausius statement.

PERPETUAL MOTION MACHINES

Perpetual-motion machine - any device that violates the first or the second law of Thermodynamics

- A device that violates the first law (by *creating* energy) is called a PMM1.
- A device that violates the second law is called a PMM2.
- Despite numerous attempts, no perpetual-motion machine is known to have worked.



A perpetual-motions machine that violates the second law (PMM2)

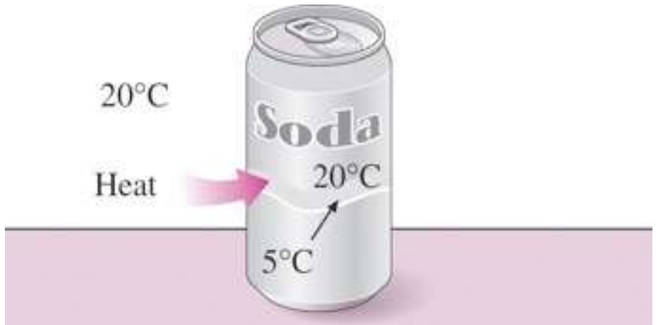
REVERSIBLE & IRREVERSIBLE PROCESSES

Reversible process - A process that can be reversed without leaving any trace on the surroundings.

Irreversible process - A process that is not reversible.

All the processes occurring in nature are irreversible.

Irreversibilities



(*a*) An irreversible heat transfer process

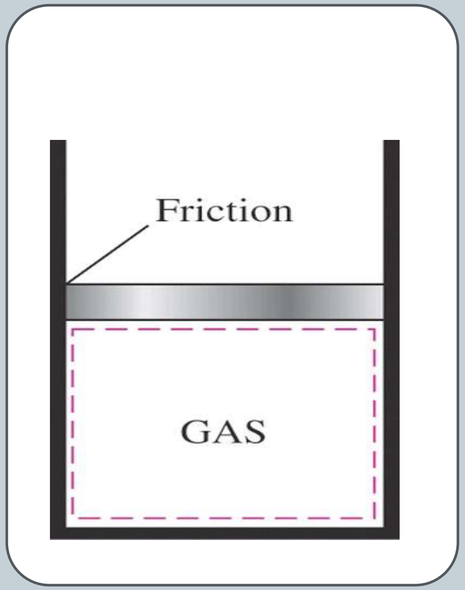


(*b*) An impossible heat transfer process

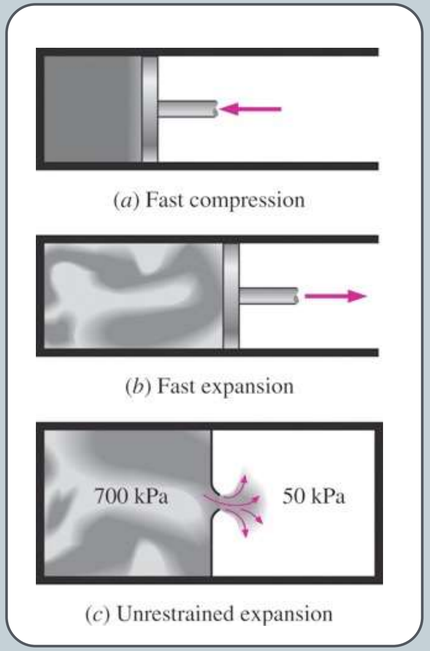
(a)Heat transfer through a temperature difference is irreversible, and
(b)the reverse process is impossible.

The factors that cause a process to be irreversible are called ***irreversibilities***

- They include friction, unrestrained expansion, mixing of two fluids, heat transfer across a finite temperature difference, electric resistance, inelastic deformation of solids, and chemical reactions.
- The presence of any of these effects renders a process irreversible.



Friction renders a process irreversible.



Irreversible compression and expansion processes.

Internally & externally reversible processes

➢**Internally reversible process**

If no irreversibilities occur within the boundaries of the system during the process.

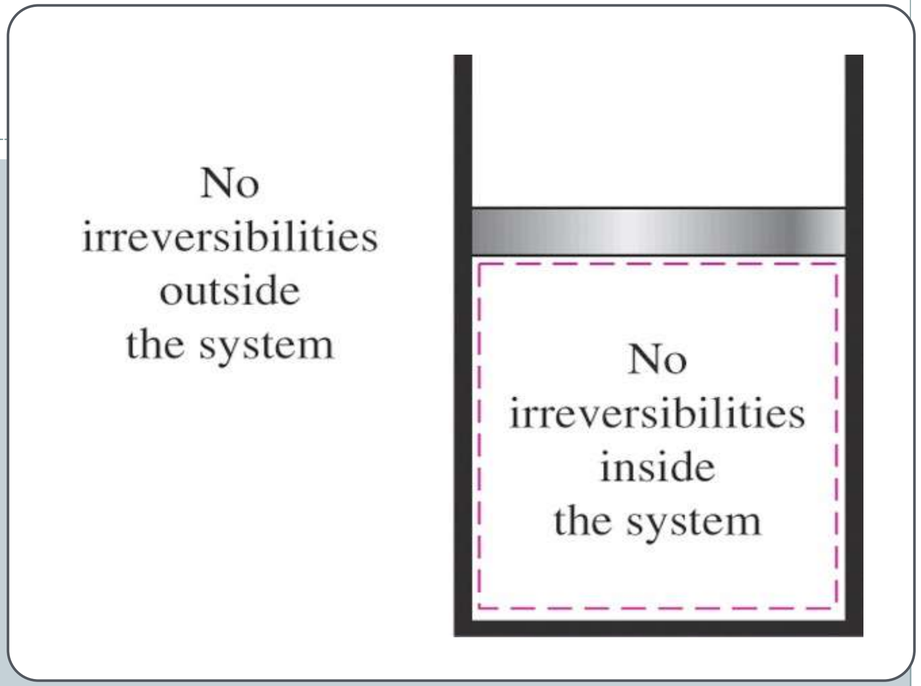
➢**Externally reversible**

If no irreversibilities occur outside the system boundaries.

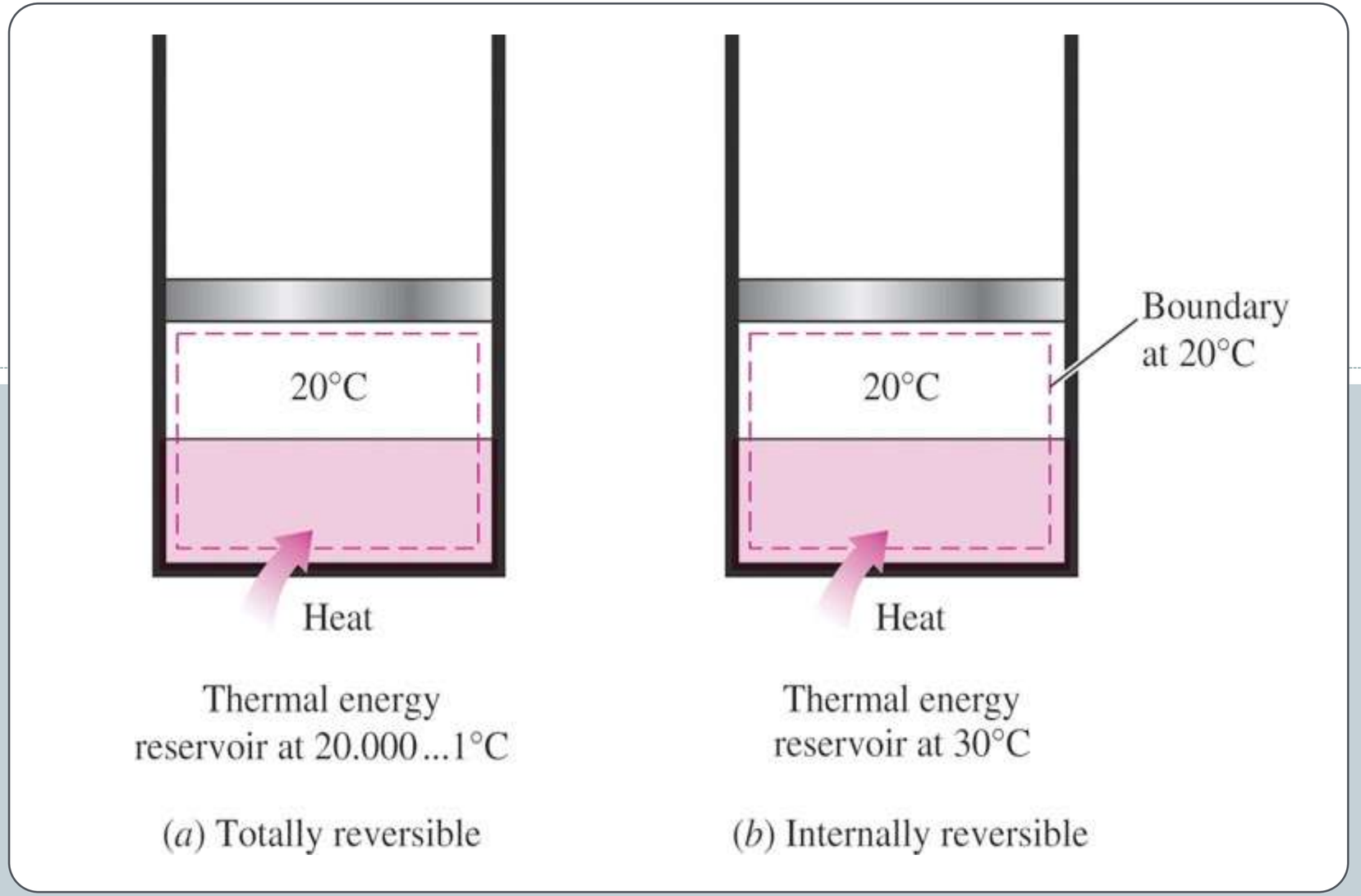
➢**Totally reversible process**

It involves no irreversibilities within the system or its surroundings.

➢A totally reversible process involves no heat transfer through a finite temp difference, no nonquasi-equilibrium changes, and no friction or other dissipative effects.



A reversible process involves no internal and external irreversibility's.



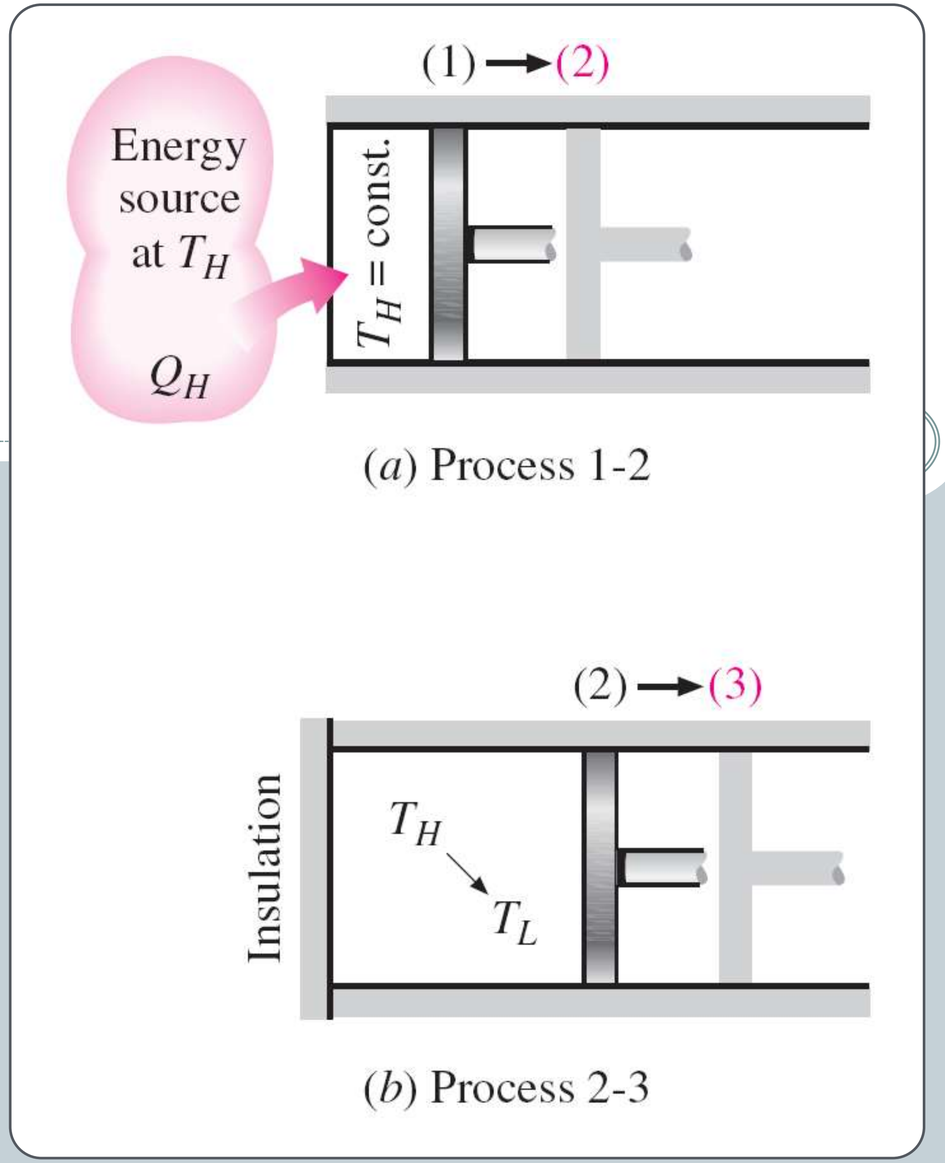
Totally and internally reversible heat transfer processes.

CARNOT CYCLE

- In an ideal heat engine, all processes are reversible and there are no losses; such an engine is the most efficient possible between the chosen reservoirs. The efficiency of an ideal heat engine is called the ***Carnot efficiency***

- Carnot's principle may be demonstrated by considering the existence of a **reversed heat engine**, which is a device, which accepts work transfer in order to transfer heat from a cold reservoir to a hot reservoir.

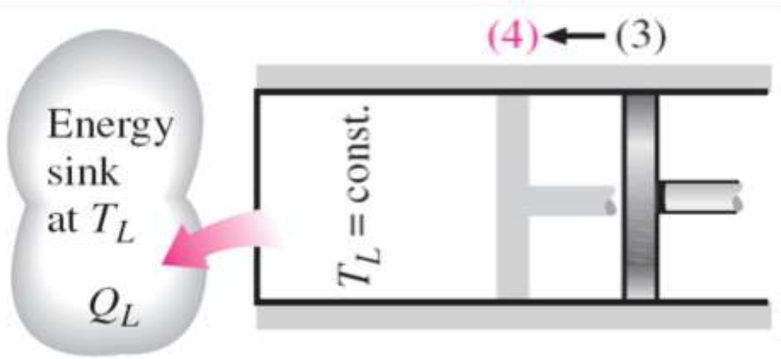
- An ideal reversed heat engine will perform reversible processes and will be the exact 'reverse' to the ideal heat engine.



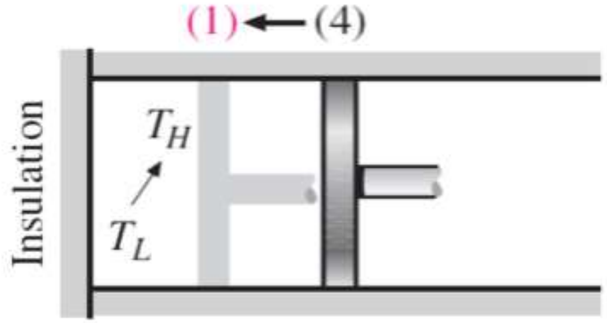
Reversible Isothermal Expansion (process 1-2, T_H = constant)

Reversible Adiabatic Expansion (process 2-3, temperature drops from T_H to T_L)

Execution of Carnot cycle in a closed system



(*c*) Process 3-4



(*d*) Process 4-1

Reversible Isothermal Compression (process 3-4, T_L = constant)

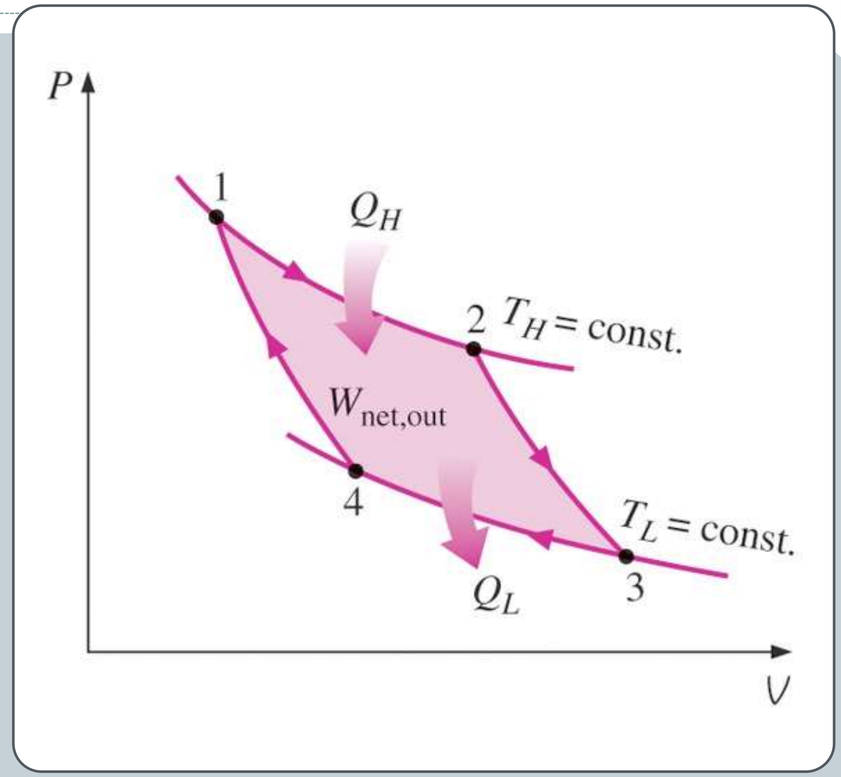
Reversible Adiabatic Compression (process 4-1, temperature rises from T_L to T_H)

Execution of Carnot cycle in a closed system

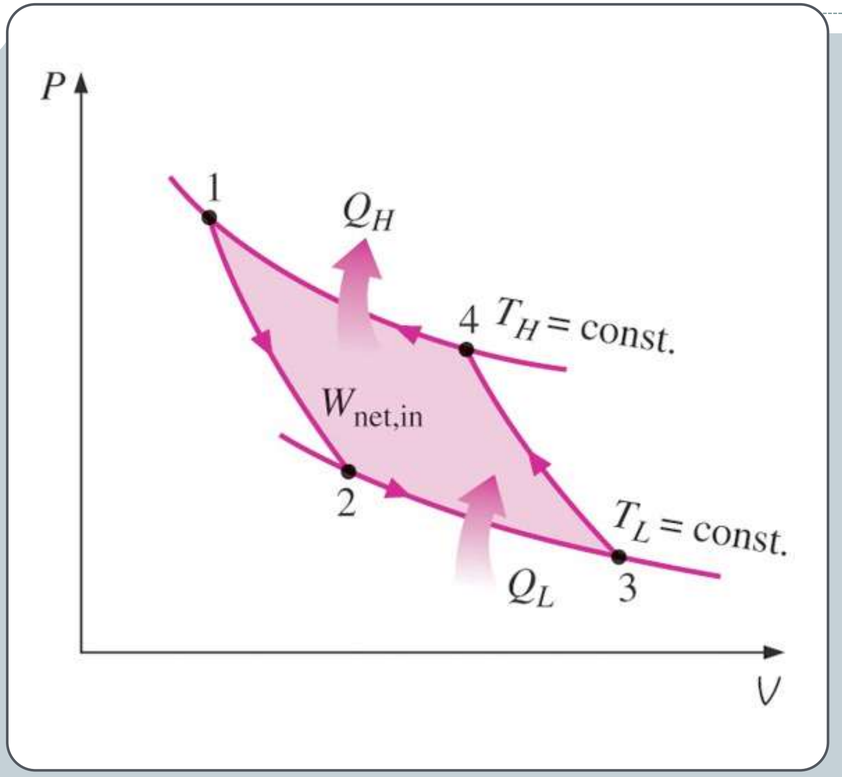
Internally & externally reversible processes

The ***Carnot heat-engine*** cycle is a totally reversible cycle.

Therefore, all the processes that comprise it can be ***reversed***, in which case it becomes the ***Carnot refrigeration cycle***.



P-V diagram of the Carnot cycle

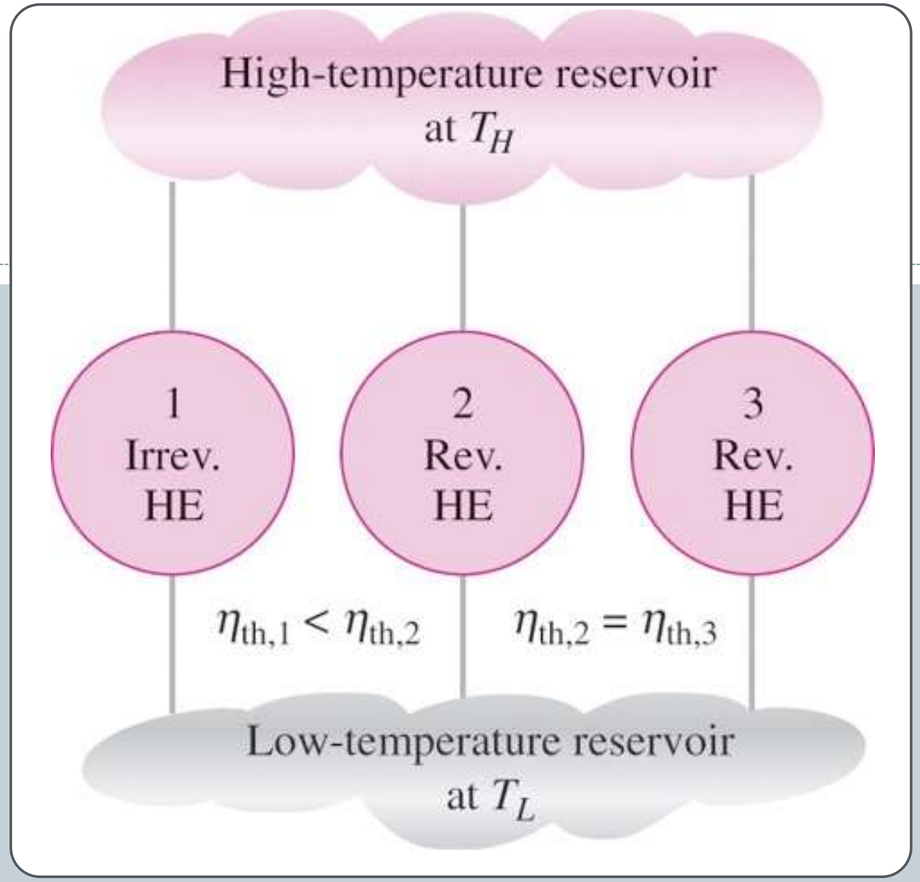


P-V diagram of the reversed Carnot cycle

Carnot's theorem

The efficiency of an irreversible heat engine is always less than the efficiency of a reversible one operating between the same two reservoirs.

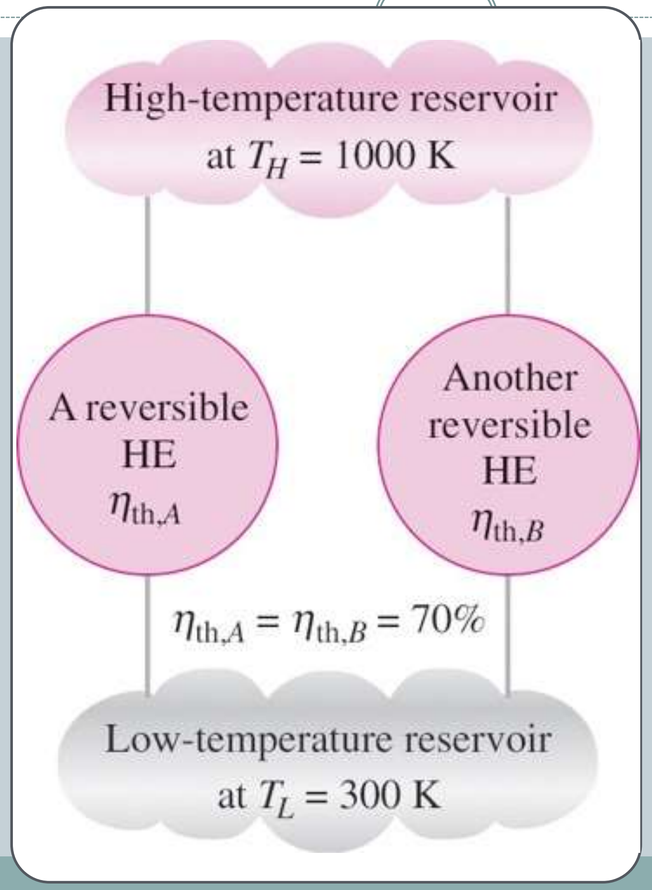
The efficiencies of all reversible heat engines operating between the same two reservoirs are the same.



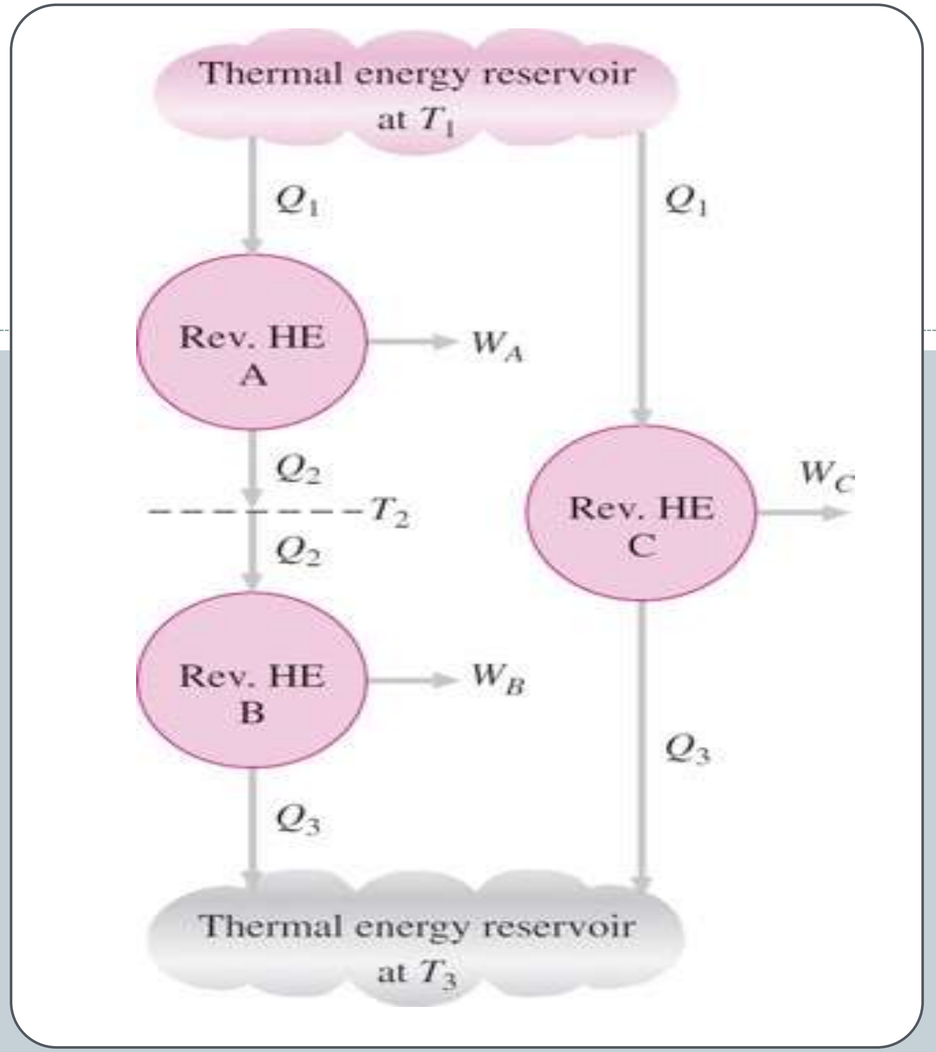
The thermodynamic temperature scale

A temperature scale that is independent of the properties of the substances that are used to measure temperature is called a **thermodynamic temperature scale**.

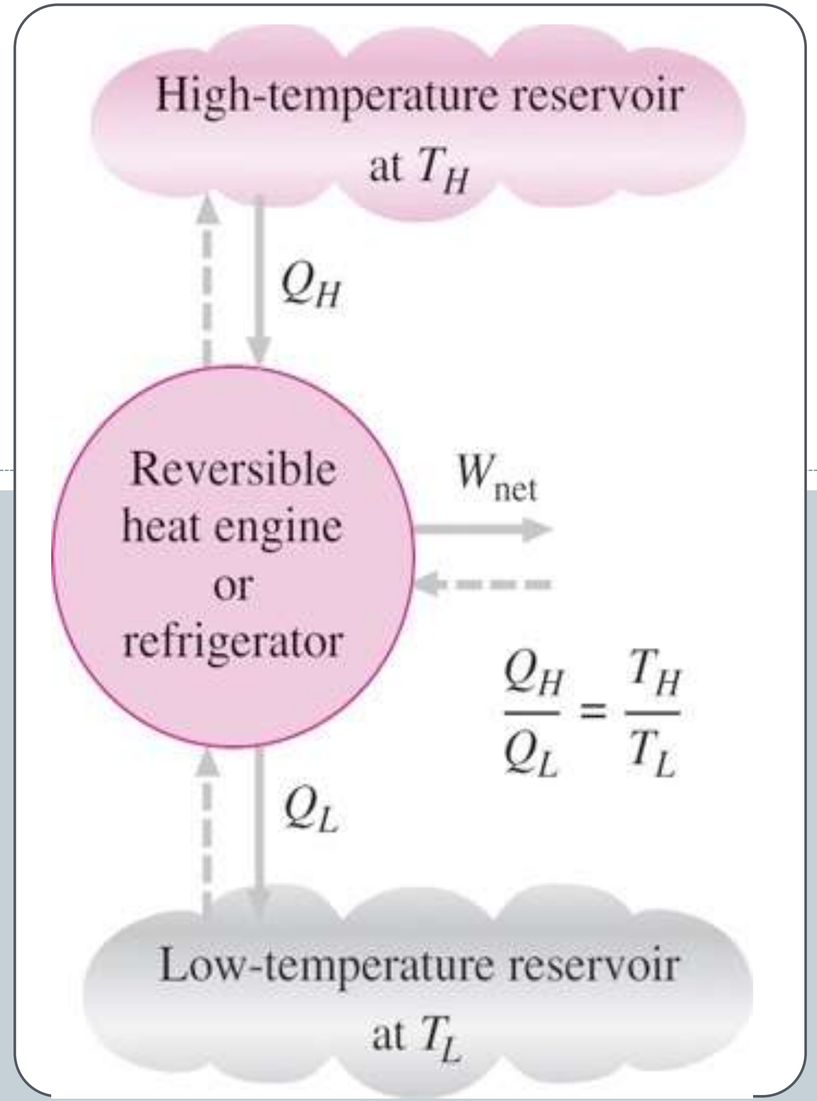
Such a temperature scale offers great conveniences in thermodynamic calculations



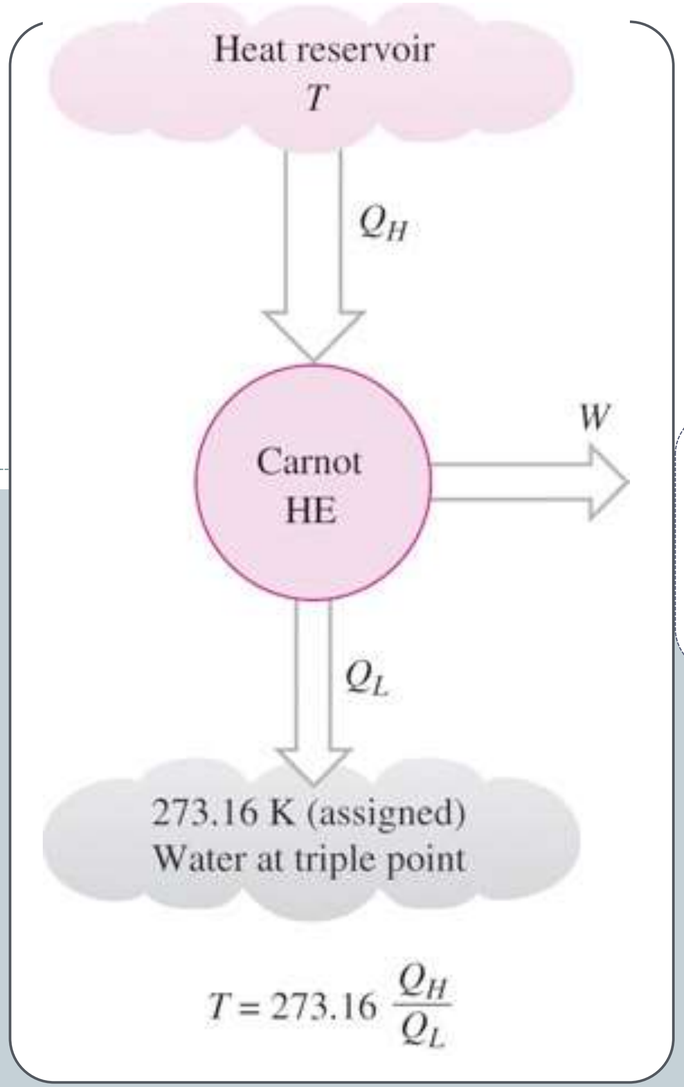
All reversible heat engines operating between the same two reservoirs have the same efficiency.



The arrangement of heat engines used to develop the thermodynamic temperature scale.



For reversible cycles, the heat transfer ratio Q_H/Q_L can be replaced by the absolute temperature ratio T_H/T_L.



A conceptual experimental setup to determine thermodynamic temperatures on the Kelvin scale by measuring heat transfers Q_H and Q_L.

$$\left(\frac{Q_H}{Q_L}\right)_{rev} = \frac{T_H}{T_L}$$

This temperature scale is called the ***Kelvin scale***, and the temperatures on this scale are called **absolute temperatures**

$$T(°C) = T(K) - 273.15$$

The Carnot Heat Engine

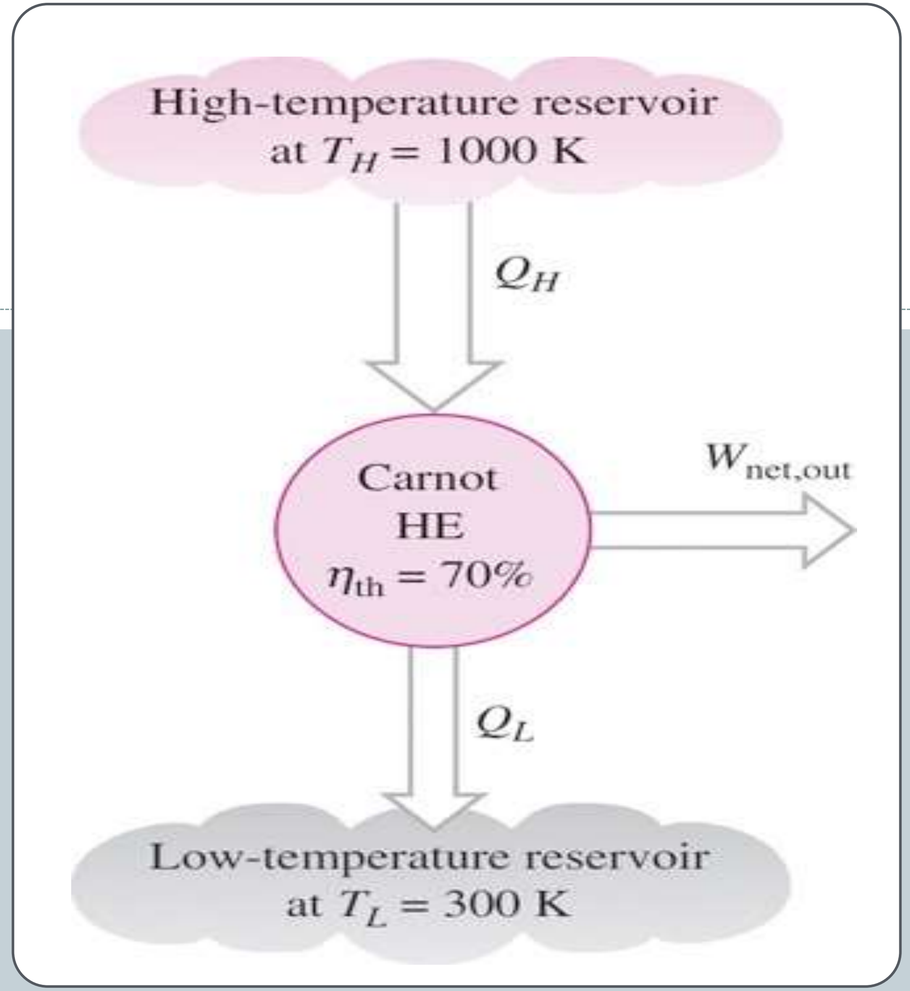
Any heat engine

$$\eta_{th} = 1 - \frac{Q_L}{Q_H}$$

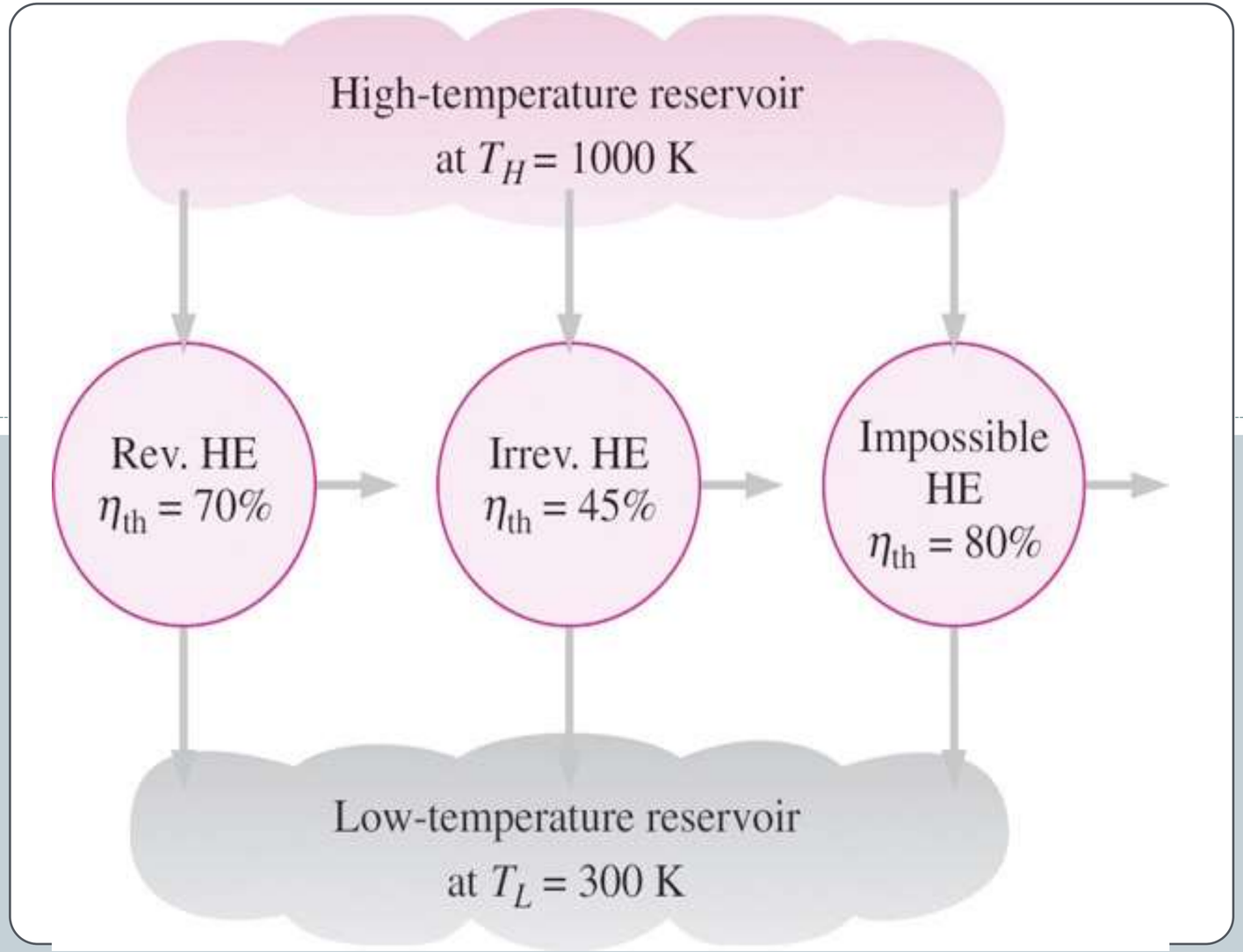
Carnot heat engine

$$\eta_{th},rev = 1 - \frac{T_L}{T_H}$$

$$\eta_{th} \begin{cases} < \eta_{th,\,rev} & \text{irreversible heat engine} \\ = \eta_{th,\,rev} & \text{reversible heat engine} \\ > \eta_{th,\,rev} & \text{impossible heat engine} \end{cases}$$

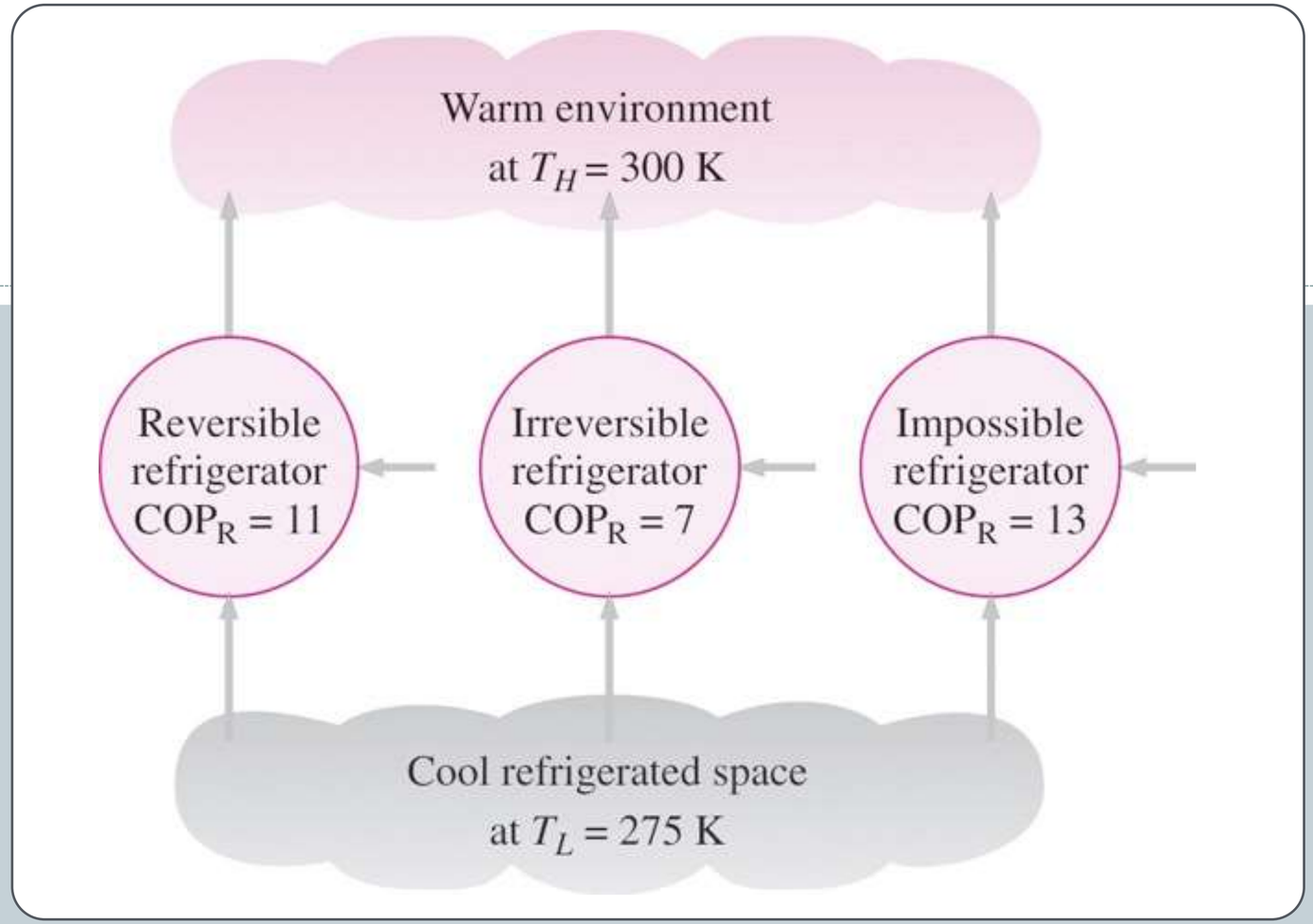


The Carnot heat engine is the most efficient of all heat engines operating between the same high and low temperature reservoirs.



No heat engine can have a higher efficiency than a reversible heat engine operating between the same high- and low-temperature reservoirs.

The Carnot refrigerator and heat pump



No refrigerator can have a higher COP than a reversible refrigerator operating between the same temperature limits.

Any refrigerator or heat pump

$$COP_R = \frac{1}{\frac{Q_H}{Q_L} - 1}$$

$$COP_{HP} = \frac{1}{1 - \frac{Q_L}{Q_H}}$$

Carnot refrigerator or heat pump

$$COP_{HP,\,rev} = \frac{1}{1 - \frac{T_L}{T_H}}$$

$$COP_{R,\,rev} = \frac{1}{\frac{T_H}{T_L} - 1}$$

Definition of Entropy

- Entropy is a thermodynamic property that measures the degree of energy dispersal or the unavailability of energy for conversion into useful work. It is also used to describe the irreversibility of a process.
- Entropy is denoted by S, and its SI unit is J/K or kJ/K. Specific entropy is denoted by s, and its SI unit is kJ/(kg·K).

$$\mathbf{dS = \delta Q_{rev} / T}$$

- where:
- dS = change in entropy
- δQ_{rev} = reversible heat transfer
- T = absolute temperature in kelvin

Significance of Entropy

- **Determines the direction of a process**: Entropy helps determine whether a process can occur naturally. For an isolated system, entropy increases in an irreversible process and remains constant in a reversible process.
- **Measures irreversibility**: Entropy generation indicates irreversibility caused by friction, mixing, heat transfer across a finite temperature difference, and other dissipative effects.
- **Second law of thermodynamics**: The total entropy of an isolated system never decreases. It increases for an irreversible process and remains constant for a reversible process.
- **Energy availability**: Entropy helps determine how much energy is unavailable for conversion into useful work.
- **Heat engine performance**: Entropy is useful in analyzing heat engines and refrigerators and in determining their maximum possible efficiency.
- **Equilibrium condition**: For an isolated system, entropy reaches its maximum value at thermodynamic equilibrium.
- **Analysis of thermodynamic processes**: Entropy is used to study expansion, compression, heating, cooling, and phase change.

Entropy Change for Different Processes

- **Reversible adiabatic process:** Entropy remains constant; this is called an isentropic process.
- **Irreversible process:** Entropy is generated.
- **Isolated system:** Entropy never decreases.